Knowledge Graphs Data Quality at Scale

Challenges, Solutions, and Applications

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[1]University of Padua

[2]Amazon

Work does not relate to O. Alonso's position at Amazon.

CIKM 2026 Tutorial · Rome, Italy

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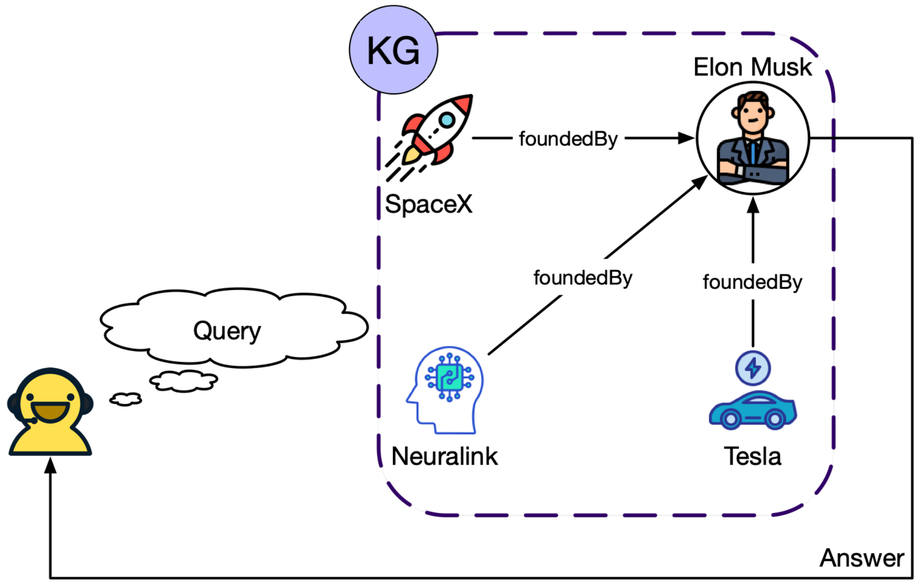
https://hereditary-project.eu/

Knowledge Graphs: Downstream Tasks

Question answering with explanation paths

KGs serve search, recommendation, and question answering [1–3]

- KG data can be used to provide answers to factual queries
  - e.g., Who is the founder of Tesla, SpaceX, and Neuralink?
- KG data can be used to provide explanations about provided answers
  - Paths from users' (entity) queries to target (entity/attribute) answers



[1] Reinanda et al. Knowledge Graphs: An Information Retrieval Perspective. Found. Trends Inf. Retr. 2020 · [2] Hogan et al. Knowledge Graphs. Morgan & Claypool 2021
[3] Weikum et al. Machine Knowledge. Found. Trends Databases 2021

Why KG Data Quality Matters

Informed use and correction mechanisms

Knowing the accuracy of KGs is fundamental for their **informed use** and to **activate correction mechanisms:**

- KG accuracy can be considered to favor search towards validated facts [1]
- KG platforms are hampered by incorrect data: high-traffic queries on low-quality KG bits [2, 3]
- KG embeddings are sensitive to unreliable data [4]
- KG data used to train LLMs: incorrect data leads to hallucinations, biases, and systematic errors [5]

[1] Reinanda et al. Knowledge Graphs: An Information Retrieval Perspective. Found. Trends Inf. Retr. 2020 · [2] Ilyas et al. Saga. SIGMOD 2022
[3] Noy et al. Industry-scale Knowledge Graphs: Lessons and Challenges. CACM 2019 · [4] Pujara et al. Sparsity and Noise: Where Knowledge Graph Embeddings Fall Short. EMNLP 2017
[5] Huang et al. A Survey on Hallucination in Large Language Models. ACM TOIS 2025

Tutorial Roadmap

Three stages of the KG lifecycle

Part 1 · KG construction and data quality challenges

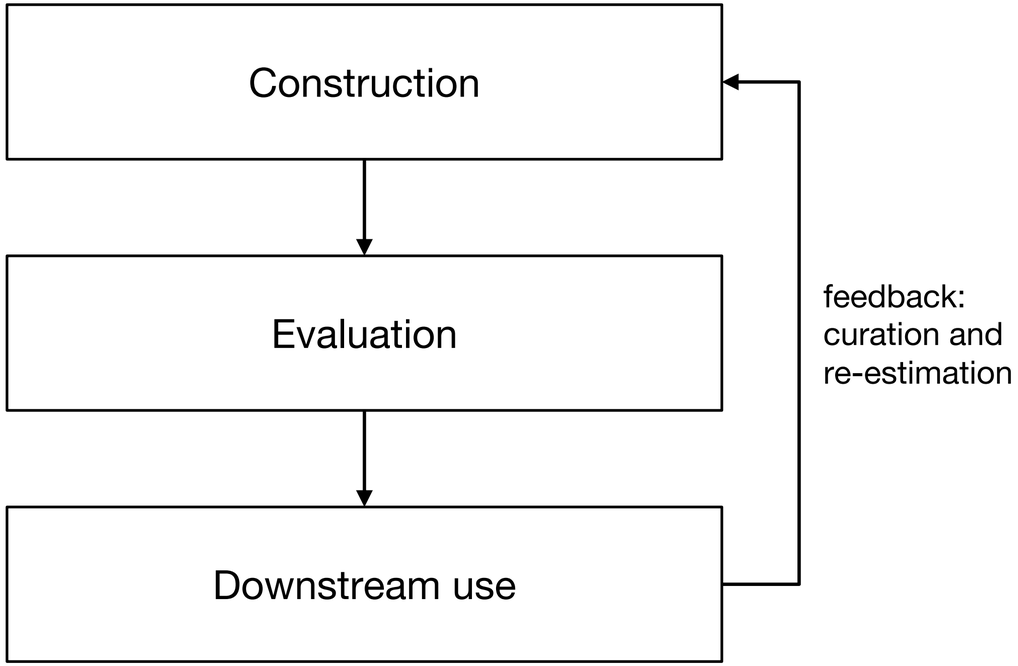
1.1 Constructing KGs at scale · 1.2 Data quality challenges of evolving KGs

Part 2 · Evaluation methodologies

2.1 Sampling · 2.2 Intervals · 2.3 Utility and budgets · 2.4 LLMs

Part 3 · Downstream applications and open problems

3.1 Veracity-driven entity search · 3.2 Open problems



Tutorial Material

Slides, code, data, and reading list in one repository

github.com/KGAccuracyEval/kg-quality-tutorial

README: outline, slides, datasets, code and data links, annotated reading list

Part 1: KG Construction and Data Quality Challenges

1.1 Constructing KGs at scale

1.2 From construction to evaluation: data quality challenges

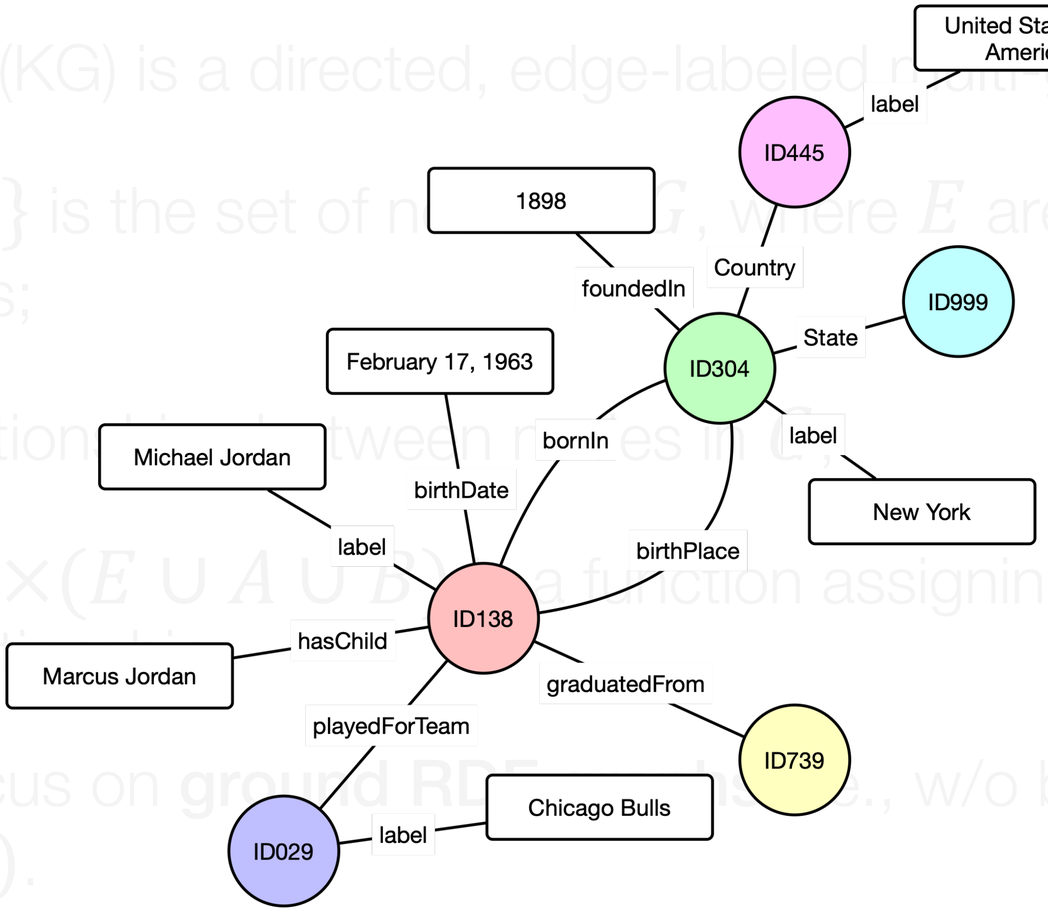
Knowledge Graphs: Preliminaries (1/2)

Entities, attributes, and relationships

A Knowledge Graph (KG) is a directed, edge-labeled multi-graph $G = (V, R, \eta)$:

- $V = \{E \cup A \cup B\}$ is the set of nodes in G, where E are entities, A attributes and B blank nodes;
- R is the set of relation labels between nodes in G;
- $\eta: R \to (E \cup B) \times (E \cup A \cup B)$ is a function assigning an ordered pair of nodes to each relation label.

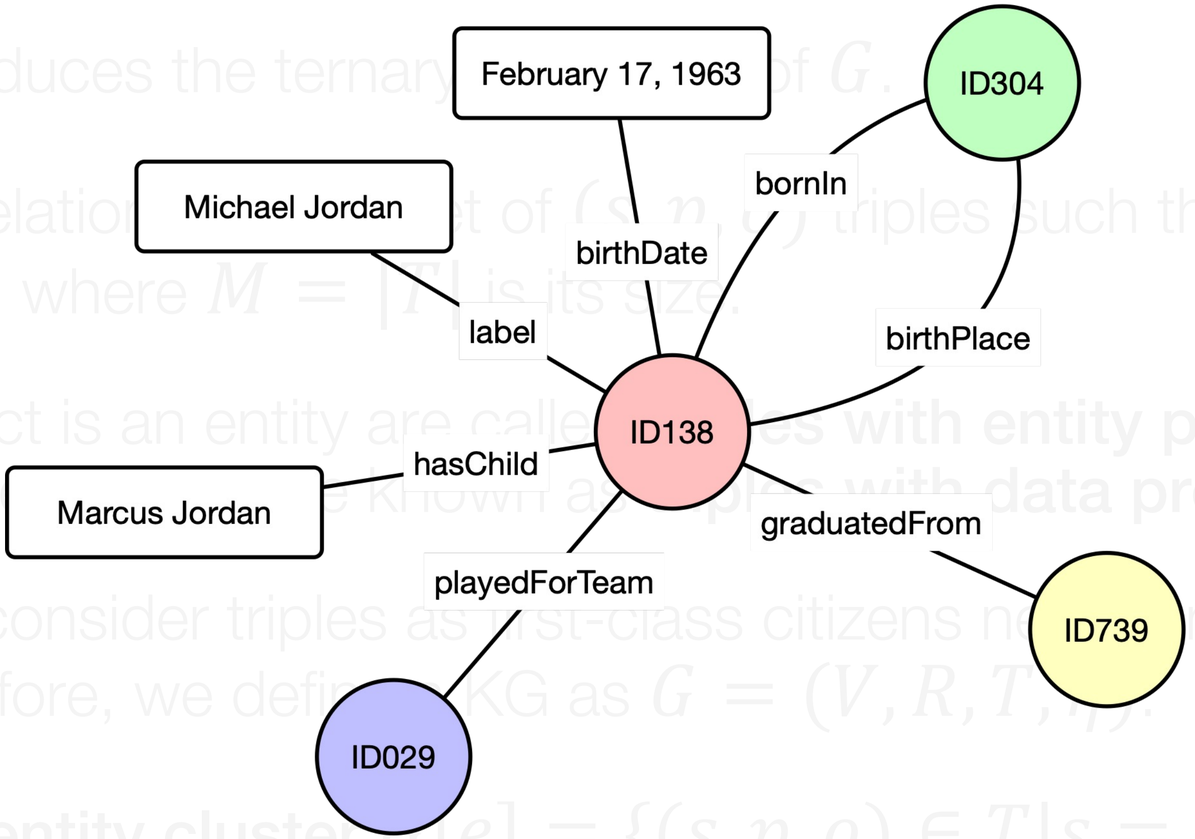
In this tutorial, we focus on **ground RDF** graphs (i.e., w/o blank nodes), hence $\eta: R \to E \times (E \cup A)$.



Bonifati et al. Querying Graphs. Morgan & Claypool 2018

Knowledge Graphs: Preliminaries (2/2)

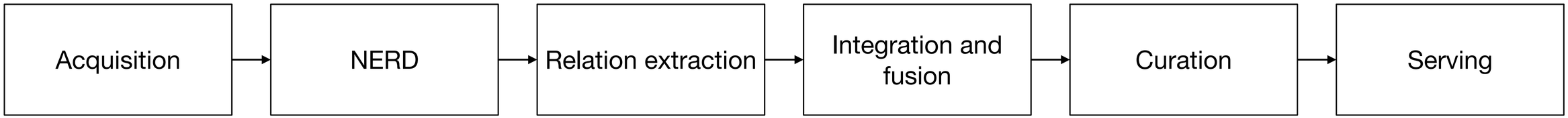
Triples and entity clusters



Building KGs: The Landscape

From encyclopedic KGs to industrial platforms

- Encyclopedic KGs from Wikipedia and collaborative editing: DBpedia, YAGO, Wikidata [1–3]
- Text-derived KGs: entity recognition and disambiguation (NERD), relation extraction, fusion [4]
- Industrial KGs: continuous construction and serving, human curation in the loop [5–7]
- Common stages, from raw sources to served facts:

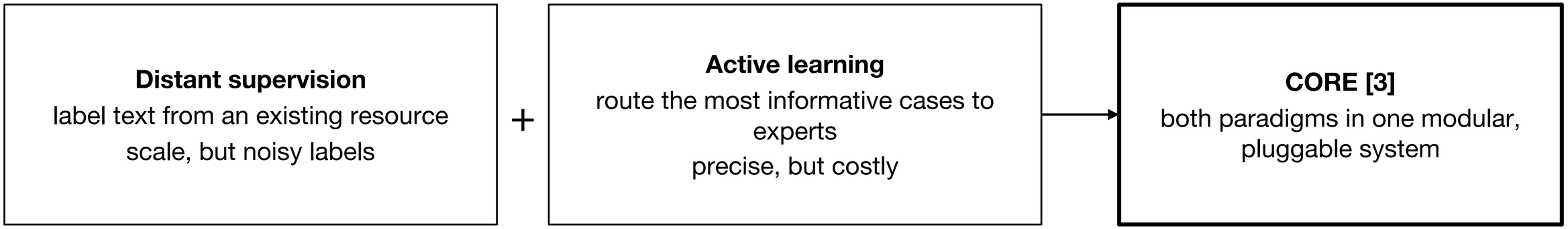


[1] Auer et al. DBpedia: A Nucleus for a Web of Open Data. ISWC 2007 · [2] Suchanek et al. YAGO 4.5. SIGIR 2024 · [3] Vrandečić and Krötzsch. Wikidata. CACM 2014
[4] Weikum et al. Machine Knowledge. Found. Trends Databases 2021 · [5] Noy et al. Industry-scale Knowledge Graphs: Lessons and Challenges. CACM 2019
[6] Ilyas et al. Saga. SIGMOD 2022 · [7] Deshpande et al. Building, Maintaining, and Using Knowledge Bases. SIGMOD 2013

Construction Under Limited Labels

The supervision bottleneck

- NERD and relation extraction need labeled training data
- Expert curation is costly and is priced per fact [1, 2]



[1] Paulheim. How much is a Triple? ISWC 2018 (Posters) · [2] Karp. How much does curation cost? Database 2016
[3] Marchesin et al. Building a Large Gene Expression-Cancer Knowledge Base with Limited Human Annotations. Database 2023

Case Study: Gene Expression and Cancer

Facts as multi-aspect relationships

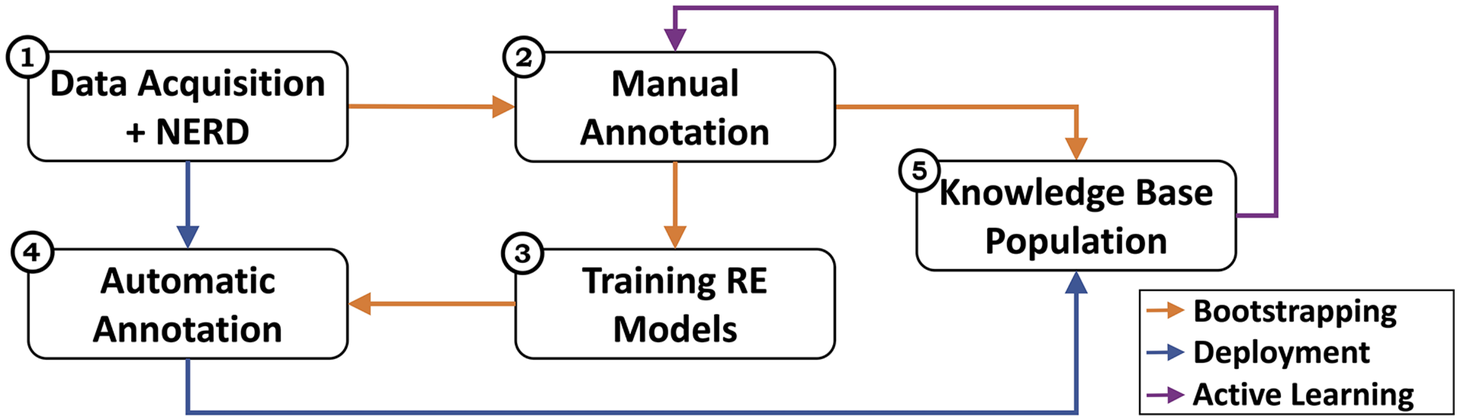
- Goal: harvest fine-grained gene expression-cancer associations from the biomedical literature
- Three aspects per gene-cancer pair: change of gene expression (CGE), change of cancer status (CCS), gene-cancer interaction (GCI)
- Inference rules map aspect tuples to gene roles:

Rule	CGE	CCS	GCI	Gene class
1	up	progression	causality	oncogene
2	up	regression	causality	tumor suppressor gene
3	down	regression	causality	oncogene
4	down	progression	causality	tumor suppressor gene
5	up | down	progression | regression	observation	biomarker

Marchesin et al. Building a Large Gene Expression-Cancer Knowledge Base with Limited Human Annotations. Database 2023

CORE: Architecture

Five modules, three workflows



(1) Data acquisition and NERD (PubTator)

(2) Manual annotation of seed sentences

(3) Relation extraction training

(4) Large-scale automatic annotation

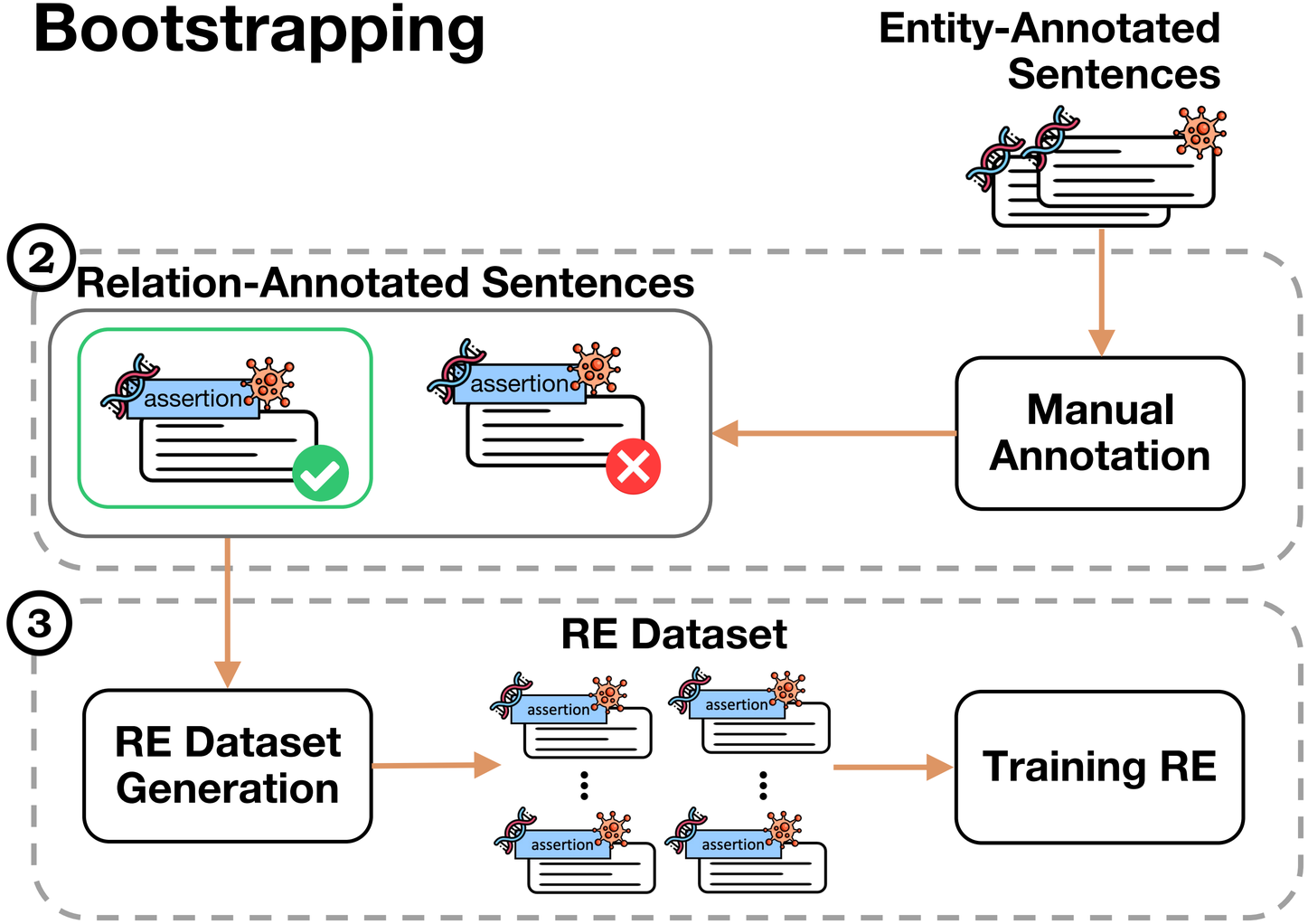
(5) KG population: knowledge enrichment and reliability testing

Workflows: bootstrapping (orange), deployment (blue), active learning (purple)

Marchesin et al. Building a Large Gene Expression-Cancer Knowledge Base with Limited Human Annotations. Database 2023

CORE: Three Workflows

Bootstrapping



Who acts: a clinical expert annotates seed sentences (CGE, CCS, GCI)

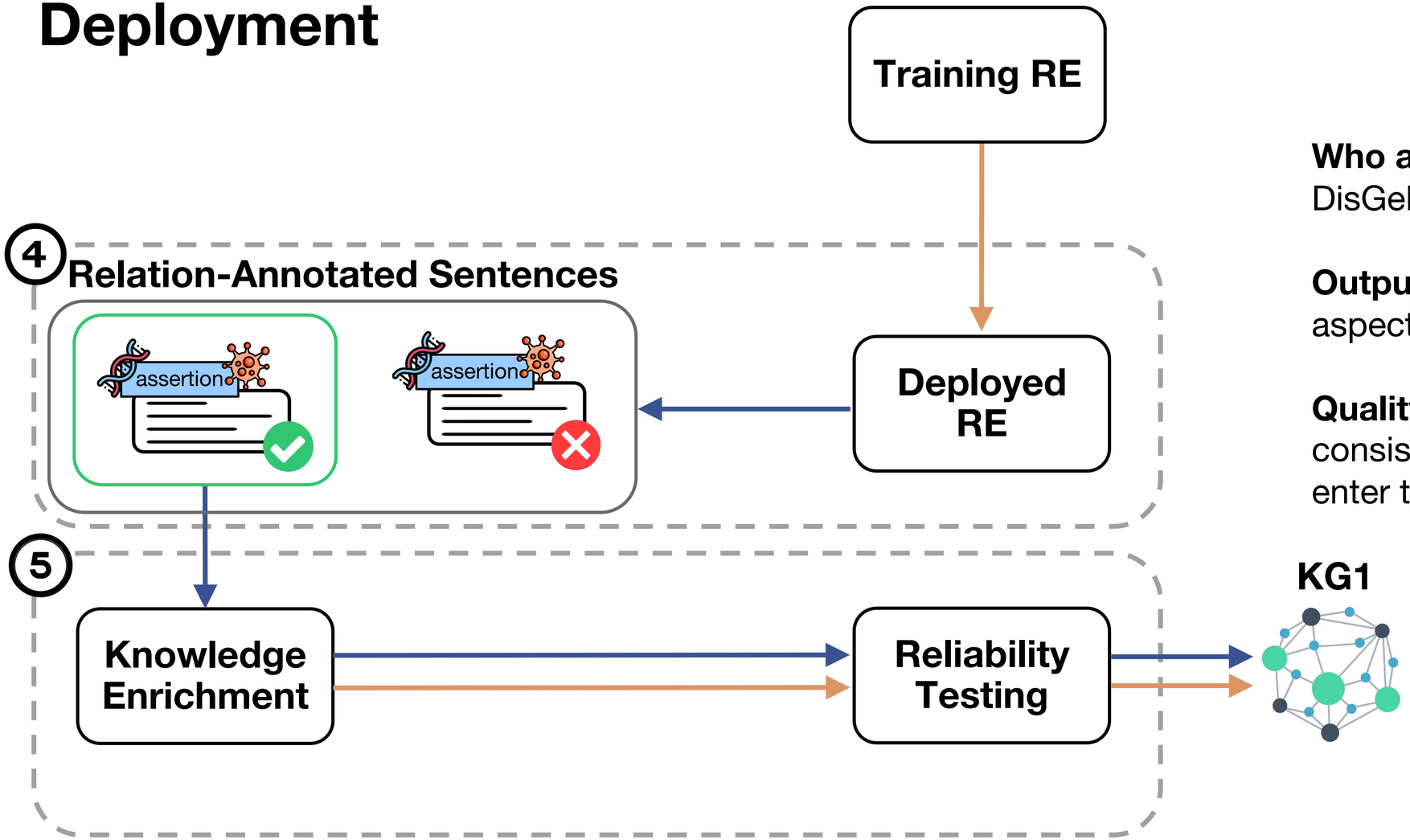
Output: one SciBERT-based extractor per aspect; the first KG version, KG0

Quality control: expert judgment on every seed sentence

Marchesin et al. Building a Large Gene Expression-Cancer Knowledge Base with Limited Human Annotations. Database 2023

CORE: Three Workflows

Deployment



Who acts: the trained extractors, on DisGeNET and PubMed sentences

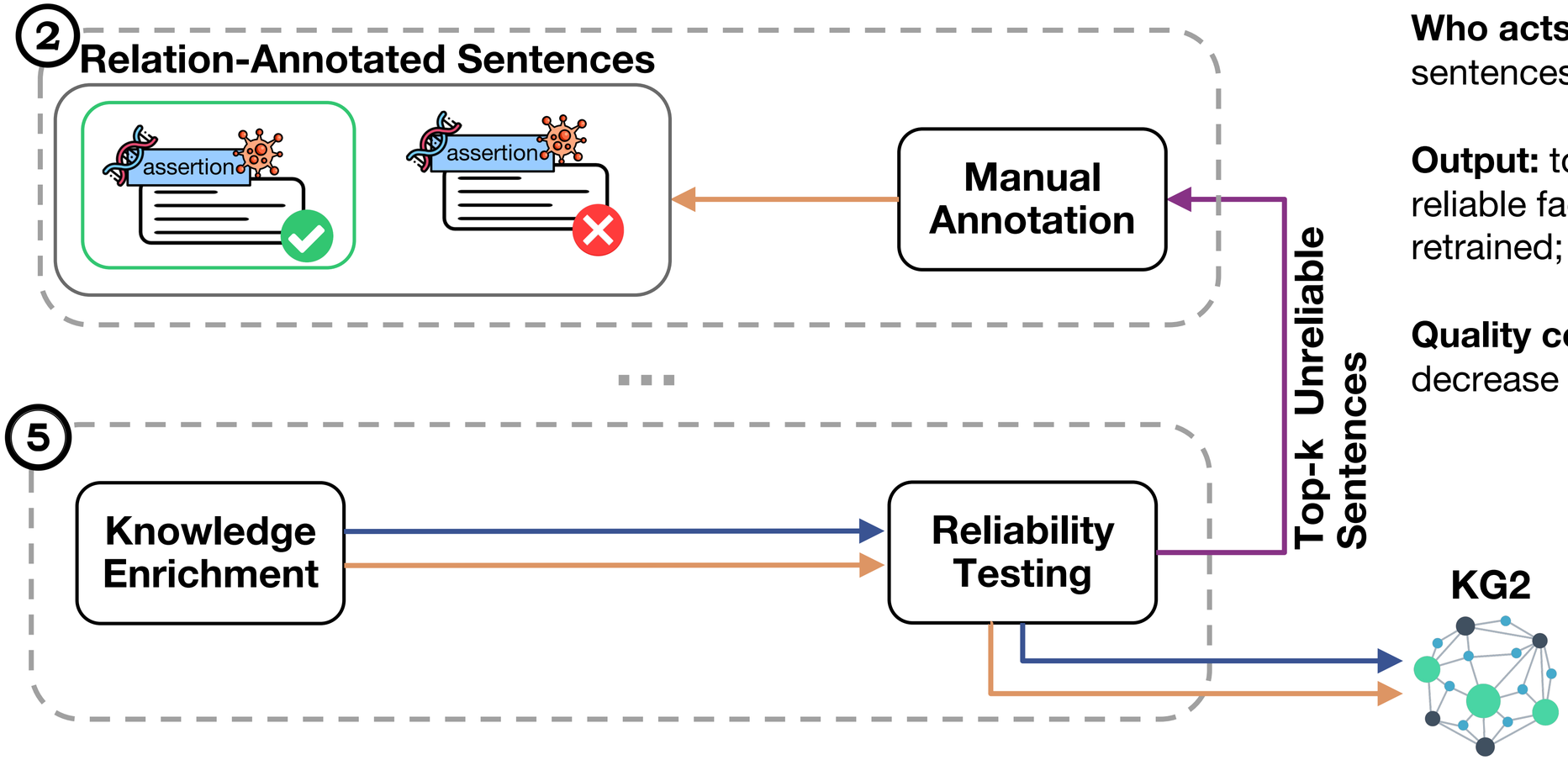
Output: facts with a probability per aspect value

Quality control: sufficiency and consistency tests; only reliable facts enter the KG

Marchesin et al. Building a Large Gene Expression-Cancer Knowledge Base with Limited Human Annotations. Database 2023

CORE: Three Workflows

Active learning



Who acts: the expert again, on sentences the system selects

Output: top-k sentences of the least reliable facts re-annotated; extractors retrained; new KG version

Quality control: unreliable facts decrease across versions

Marchesin et al. Building a Large Gene Expression-Cancer Knowledge Base with Limited Human Annotations. Database 2023

CORE in Numbers

Four KG versions from limited expert input

Version	Manual input (cumulative)	In the KG: manual	In the KG: automatic	Facts	Genes	Cancer types
KG0	1,052	655	0	512	329	98
KG1	1,052	585	96,531	71,554	9,483	1,479
KG2	1,302	605	95,282	89,999	9,981	1,554
KG3	1,551	592	435,283	153,016	21,005	1,665

Sentence counts are those supporting reliable facts, i.e., facts that pass the sufficiency and consistency tests; manual input grows with the sentences selected by active learning. The share of (manual) input sentences that ends in the KG falls from 62% (KG0) to 14% (KG3)

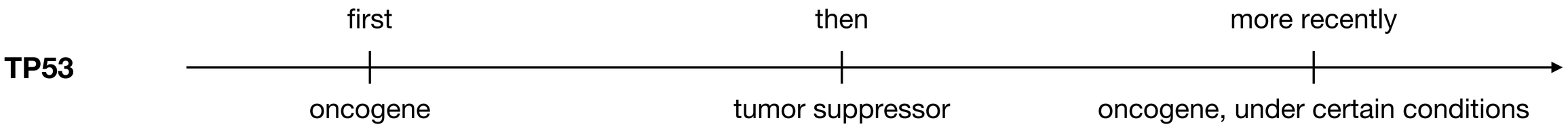
- KG3 covers 21,005 genes, about 70% of the ~30,000 estimated human genes
- Relation extractors: F1 0.81–0.88 per aspect
- Reconstruction of 9,636 held-out BioXpress sentences: F1 0.96–0.98; pattern matching (DEXTER) reaches F1 0.29 on heterogeneous sentences
- KG and SPARQL endpoint: zenodo.org/record/7577127

Marchesin et al. Building a Large Gene Expression-Cancer Knowledge Base with Limited Human Annotations. Database 2023

Evidence Is Uncertain and Evolves

Conflicting sentences and changing roles

- Sentences about the same pair support different roles: for BRAF in colorectal cancer, one sentence gives oncogene, another biomarker
- CORE stores a probability per aspect value and keeps the supporting sentences as provenance
- Roles change with scientific understanding, as for TP53:

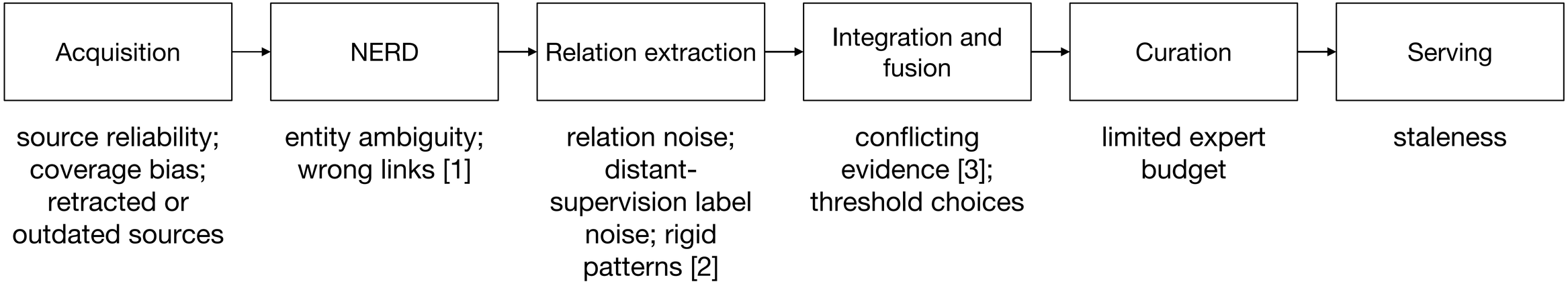


A fact verified today can be outdated tomorrow (1.2)

Marchesin et al. Building a Large Gene Expression-Cancer Knowledge Base with Limited Human Annotations. Database 2023

Where Quality Issues Enter

Failure points along the construction pipeline



Coverage skew across stages: head entities have more facts and evidence than tail entities [4]

[1] Qi et al. Evaluating KG Accuracy Powered by Optimized Human-Machine Collaboration. KDD 2022
[2] Marchesin et al. Building a Large Gene Expression-Cancer Knowledge Base with Limited Human Annotations. Database 2023
[3] Weikum et al. Machine Knowledge. Found. Trends Databases 2021 · [4] Sun et al. Head-to-Tail: How Knowledgeable are Large Language Models? NAACL 2024

Lessons from Industrial KGs

Construction is a continuous process

- An imperfect KG is still useful; maintenance matters as facts evolve [1, 2]
- Industry-scale KGs integrate many sources and serve many applications [3]
- Construction and serving run continuously; on-demand accuracy evaluation is a platform feature [4, 5]
- Human curation stays in the loop, under a budget

[1] Deshpande et al. Building, Maintaining, and Using Knowledge Bases. SIGMOD 2013 · [2] Marchesin et al. Gene Expression-Cancer Knowledge Base. Database 2023
[3] Noy et al. Industry-scale Knowledge Graphs: Lessons and Challenges. CACM 2019 · [4] Ilyas et al. Saga. SIGMOD 2022
[5] Ilyas et al. Growing and Serving Large Open-domain Knowledge Graphs. SIGMOD Companion 2023

Take-Aways: 1.1

Constructing KGs at scale



Construction at scale trades expert labels for automation; quality depends on how the two are combined



Uncertainty enters at every stage: linking, extraction, fusion



Probabilities and provenance make later evaluation possible



Facts evolve: construction-time checks do not settle correctness

Next: what happens once the KG is deployed (1.2)

Part 1: KG Construction and Data Quality Challenges

1.1 Constructing KGs at scale

1.2 From construction to evaluation: data quality challenges

After Deployment, Facts Change

Three scenarios from evolving KGs

E1 · Gene role

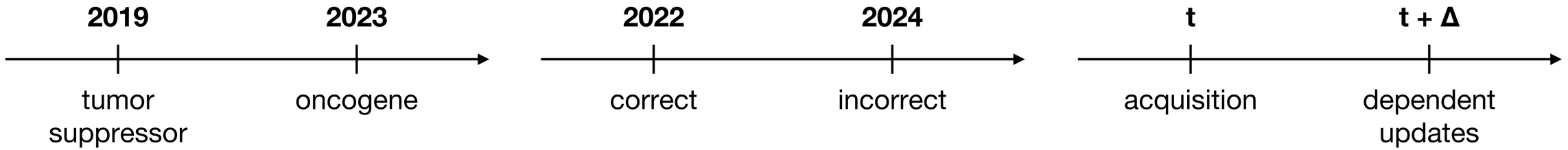
Gene X is a tumor suppressor (2019 evidence), reclassified as an oncogene in 2023

E2 · Athlete transfer

plays-for(A, Team1) is correct in 2022 and incorrect in 2024

E3 · Acquisition

An acquisition triggers dependent updates: subsidiaries, executive roles, stock identifiers



Correctness depends on when a statement holds, not only on source reliability [1, 2]

[1] Marchesin et al. Cost-Effective Quality Evaluation for Evolving Knowledge Graphs. ACM J. Data Inf. Qual. 2026
[2] Polleres et al. How Does Knowledge Evolve in Open Knowledge Graphs? TGDK 2023

Quality Dimensions and Scope

This tutorial focuses on accuracy

- Data quality is multi-dimensional: accuracy, completeness, consistency, timeliness [1–3]
- Accuracy: semantic validity of a triple, i.e., whether the statement it expresses is correct [4]
- Binary judgment per triple: correct or incorrect, whatever the error type [4, 5]
- Parts 2 and 3 estimate and use accuracy; the other dimensions are out of scope

[1] Wang and Strong. Beyond Accuracy: What Data Quality Means to Data Consumers. J. Manag. Inf. Syst. 1996
[2] Zaveri et al. Quality Assessment for Linked Data: A Survey. Semantic Web 2016 · [3] Fan. Data Quality: From Theory to Practice. SIGMOD Rec. 2015
[4] Marchesin et al. Utility-Oriented KG Accuracy Estimation with Limited Annotations. HCOMP 2024
[5] Esteves et al. Toward Veracity Assessment in RDF Knowledge Bases. ACM J. Data Inf. Qual. 2018

Structural Checks Are Not Enough

What constraint validation misses

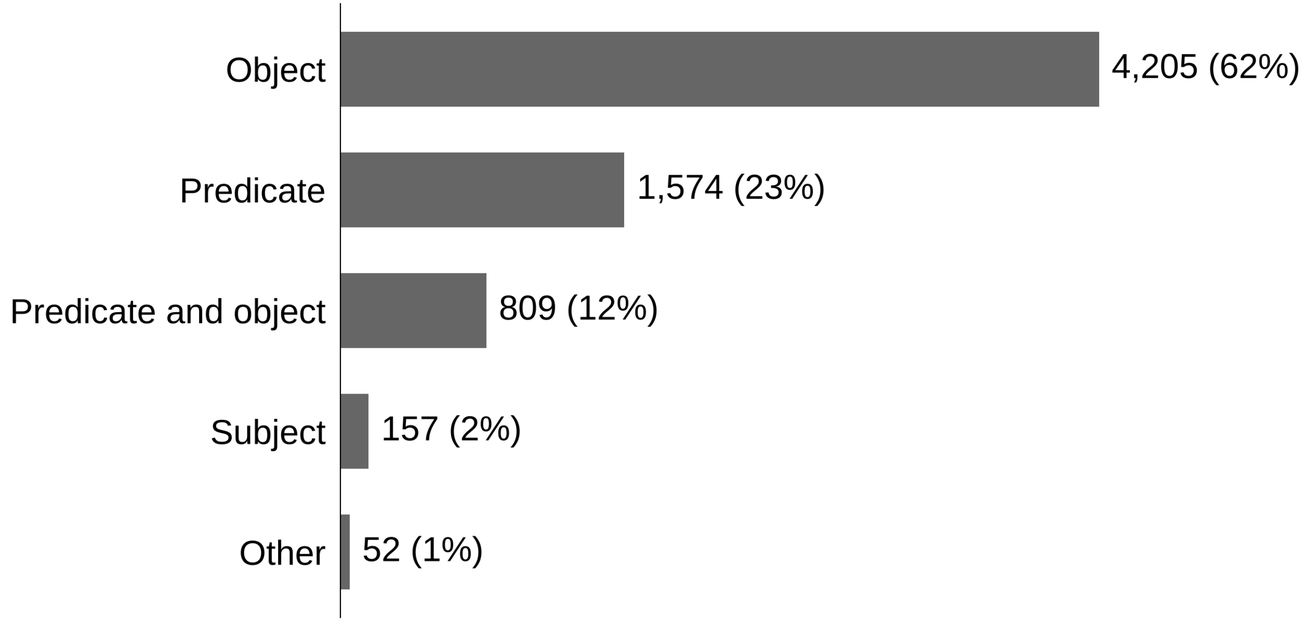
- Constraint languages such as SHACL capture structural, constraint-based checks [1]
- A stale fact is well-formed: plays-for(A, Team1) passes every shape in 2024
- A semantic error is well-typed: a wrong but plausible object value
- Temporal coherence and domain semantics must be made explicit to become testable [1, 2]
- Large-scale expert re-verification is the fallback, and it is expensive (next slides)

[1] Marchesin et al. Cost-Effective Quality Evaluation for Evolving Knowledge Graphs. ACM J. Data Inf. Qual. 2026
[2] Paulheim. Knowledge Graph Refinement: A Survey. Semantic Web 2017

Where Errors Sit in a Real KG

Human audit of DBpedia 2015-10

6,797 of 37,546 annotations reported an incorrect triple [1]
Error location:



Two experts agreed on 77% of honey-pot triples (Cohen's κ = 0.51): judging correctness is hard

Triple-level estimators assume entity linking is infallible; linking errors are a distinct class [2]

[1] Marchesin et al. Utility-Oriented KG Accuracy Estimation with Limited Annotations: A Case Study on DBpedia. HCOMP 2024
[2] Qi et al. Evaluating KG Accuracy Powered by Optimized Human-Machine Collaboration. KDD 2022

≈ $1 billion

One expert judgment for each of the 170M A-Box triples of DBpedia 2015-10 at $5.71 per triple [1–3], for a single snapshot

[1] Marchesin et al. Utility-Oriented KG Accuracy Estimation with Limited Annotations. HCOMP 2024 (170M A-Box triples after filtering)
[2] Marchesin et al. LLMs as Stratification Signals for KG Accuracy Evaluation. PVLDB 2026 ($5.71 per triple per annotator)
[3] Paulheim. How much is a Triple? ISWC 2018 (Posters)

The Cost Structure of Verification

Why full re-verification does not scale

- Each triple needs entity identification and evidence collection from external sources: up to several minutes per claim [1, 2]
- Standard setups use three or more annotators per triple [3, 4]
- Every update batch reopens the question for the facts it touches [5]
- Verification value is not uniform: frequently queried facts matter more to applications than unused ones [6]

[1] Gao et al. Efficient Knowledge Graph Accuracy Evaluation. PVLDB 2019 · [2] Shami et al. Benchmarking Large Language Models for Knowledge Graph Validation. EDBT 2026
[3] Marchesin et al. LLMs as Stratification Signals for KG Accuracy Evaluation. PVLDB 2026
[4] Marchesin and Silvello. Credible Intervals for KG Accuracy Estimation. Proc. ACM Manag. Data (SIGMOD) 2025
[5] Marchesin et al. Cost-Effective Quality Evaluation for Evolving Knowledge Graphs. ACM J. Data Inf. Qual. 2026
[6] Marchesin et al. Utility-Oriented KG Accuracy Estimation with Limited Annotations. HCOMP 2024

Common Practice: Ad Hoc Samples

Samples without statistical guarantees

Objective: evaluate the accuracy of KGs to make an informed use of them.

Solution (common practice): carry out manual annotations on a sample of the KG.



Sample size is decided beforehand: is it large enough, unbiased, reproducible?



Confidence intervals (CIs) are not reported: how close is the estimate to the true value?

No guarantees of representativeness of the sample w.r.t. the entire KG. **[1, 2]**

[1] Ojha and Talukdar. KGEval. EMNLP 2017 · [2] Gao et al. Efficient Knowledge Graph Accuracy Evaluation. PVLDB 2019

Automated Validation: Trade-Offs

No validator is both cheap and reliable

Approach	Strength	Limitation
Expert annotation [1]	accurate	expensive and slow
Rule- and constraint-based [1, 2]	scalable	misses rare and complex cases
Embedding-based [3]	fast	falls short on sparse, noisy data
LLM-based	flexible, broad knowledge	unstable, hallucinations (more in 2.4)

[1] Shami et al. Benchmarking Large Language Models for Knowledge Graph Validation. EDBT 2026 · [2] Paulheim. Knowledge Graph Refinement: A Survey. Semantic Web 2017
[3] Pujara et al. Sparsity and Noise: Where Knowledge Graph Embeddings Fall Short. EMNLP 2017

The central question of the tutorial

How can we estimate the fraction of correct facts in a KG

… with controlled cost

annotation effort and budgets spent where they matter

Part 2 · 2.1, 2.3, 2.4

… with statistical reliability

intervals that keep their nominal coverage

Part 2 · 2.2

… and channel the evidence into downstream pipelines?

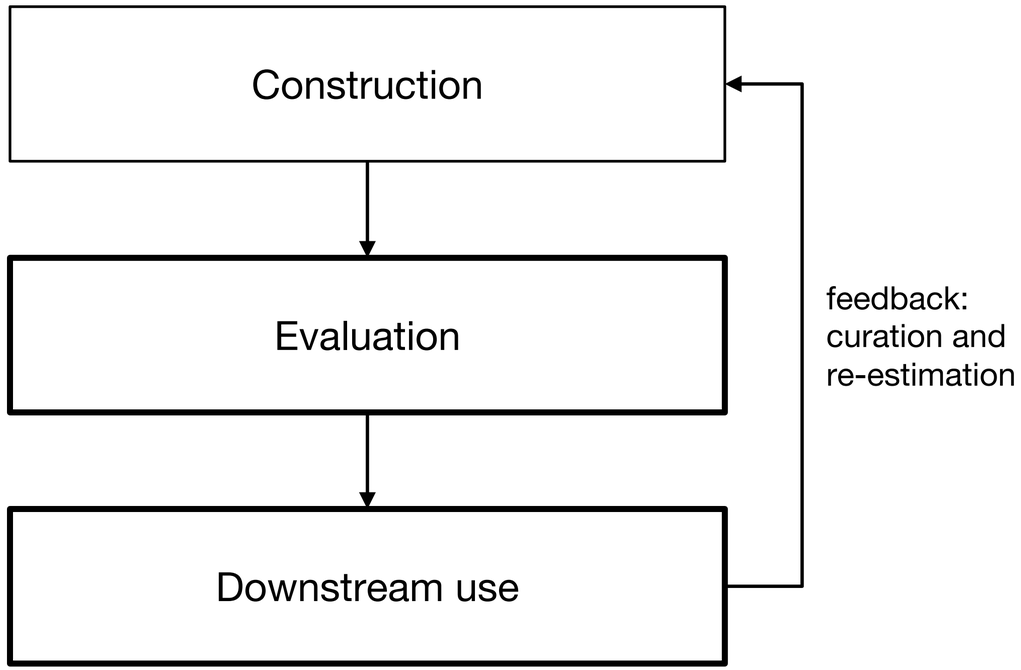
estimates consumed by search and curation

Part 3

Roadmap for Parts 2 and 3

From challenges to estimation

- Re-verifying everything is unaffordable: estimate accuracy, with guarantees, under a budget
- Which facts to sample, and at what cost? (2.1)
- How to report uncertainty correctly? (2.2)
- Where to spend a limited budget? (2.3)
- Can LLMs replace or support annotators? (2.4)
- How do estimates change entity search and help find defects? (3.1)
- What remains open, from evolving KGs to LLM applications? (3.2)



Marchesin et al. Cost-Effective Quality Evaluation for Evolving Knowledge Graphs. ACM J. Data Inf. Qual. 2026

Take-Aways: 1.2

From construction to evaluation



Evolving KGs make correctness time-dependent



Structural checks are necessary, not sufficient



Exhaustive re-verification is unaffordable: about $1 billion for one expert pass over DBpedia



Hence: estimate accuracy with guarantees under a budget

Next: Part 2, estimating accuracy under a budget

Part 2: Evaluation Methodologies

2.1 Sampling-based accuracy estimation

2.2 From Wald to Wilson and Bayesian credible intervals

2.3 Utility-oriented and budget-aware estimation

2.4 LLMs for KG quality assessment

Datasets Used in Part 2

Small human-labeled samples, synthetic KGs for scale

Dataset	Content	Facts	Accuracy μ	Labels	Used in
YAGO	sample of YAGO2: people, organizations, movies	1,386	0.99	crowd	2.1, 2.2, 2.4
NELL	sample of NELL: sports	1,860	0.91	crowd	2.1, 2.2, 2.4
DBpedia	sample of DBpedia 2015-10, broad topics	9,344	0.85	crowd + expert QC	2.2, 2.3, 2.4
FactBench	DBpedia and Freebase facts with injected errors	2,800 / 4,219	0.54 / 0.28	synthetic errors	2.2, 2.4
DisGeNET	gene–disease associations	≈ 3M	0.76 (CEM)	synthetic	2.2
YAGO-HQ / MQ / LQ	YAGO 4.5 with entity substitutions	20.4M	0.75 / 0.50 / 0.25	synthetic errors	2.4

Ojha and Talukdar. KGEval. EMNLP 2017 · Marchesin et al. Utility-Oriented KG Accuracy Estimation with Limited Annotations. HCOMP 2024 · Gerber et al. DeFacto. J. Web Semant. 2015
Piñero et al. DisGeNET. Nucleic Acids Res. 2020 · Suchanek et al. YAGO 4.5. SIGIR 2024

KG Accuracy and Its Estimation

The estimand and what an estimator must provide

$$\mu(G) = \frac{1}{M}\sum_{t \in T}\mathbf{1}(t) = \frac{\tau}{M}$$

- M triples, τ of them correct; $1(t) \in \{0, 1\}$ from manual annotation [1, 2]
- Granularity follows the downstream target: global μ(G), per predicate, per entity cluster μ(G[e])
- An estimator $\hat{\mu}$ computed on a sample must be unbiased: E[$\hat{\mu}$] = μ(G) [1]
- It must come with a 1 – α interval that quantifies sampling uncertainty

[1] Gao et al. Efficient Knowledge Graph Accuracy Evaluation. PVLDB 2019
[2] Esteves et al. Toward Veracity Assessment in RDF Knowledge Bases: An Exploratory Analysis. ACM J. Data Inf. Qual. 2018

Problem Formulation: Constrained Minimization

Minimal annotation cost, unbiased estimate, bounded MoE

Given:

- a KG G
- a sampling strategy $\mathcal{S}$
- an annotation cost function $cost(\cdot)$
- an upper bound ε for the Margin of Error (MoE) of the $1-\alpha$ CI

The problem of efficient KG accuracy evaluation can be defined as:

$$\underset{\mathcal{S}}{\text{minimize}}\ \text{cost}(\mathcal{S}(G))$$
$$\text{subject to}\ E[\hat{\mu}] = \mu(G), \text{MoE} \leq \varepsilon$$

Gao et al. Efficient Knowledge Graph Accuracy Evaluation. PVLDB 2019

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The sampling strategy draws the pool of triples that annotators audit for correctness.

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The annotation cost represents the (total) time required by annotators to audit the sample. Computational efficiency is not a direct concern, as long as the annotation time dominates.

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- an annotation cost function $cost(\cdot)$
- an upper bound ε for the Margin of Error (MoE) of the $1-\alpha$ CI

Both the upper bound ε and the significance level α are user-defined parameters. The MoE represents half the width of a CI.

The problem of efficient KG accuracy evaluation can be defined as:

$$\underset{\mathcal{S}}{\text{minimize}}\ \text{cost}(\mathcal{S}(G))$$
$$\text{subject to}\ E[\hat{\mu}] = \mu(G), \text{MoE} \leq \varepsilon$$

Gao et al. Efficient Knowledge Graph Accuracy Evaluation. PVLDB 2019

Problem Formulation: Constrained Minimization

Minimal annotation cost, unbiased estimate, bounded MoE

Given:

- a KG G
- a sampling strategy $\mathcal{S}$
- an annotation cost function $cost(\cdot)$
- an upper bound ε for the Margin of Error (MoE) of the $1-\alpha$ CI

The problem of efficient KG accuracy evaluation can be defined as:

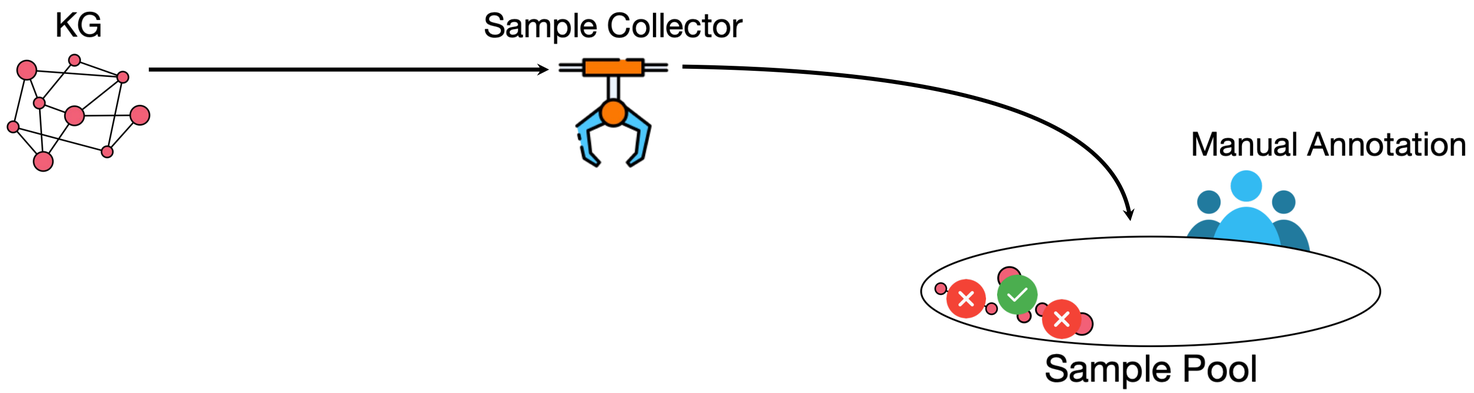
$$\underset{\mathcal{S}}{\text{minimize}}\ \text{cost}(\mathcal{S}(G))$$
$$\text{subject to}\ E[\hat{\mu}] = \mu(G), \text{MoE} \leq \varepsilon$$

- Sampling strategy $\mathcal{S}$ should minimize $\text{cost}(\cdot)$
- MoE represents the stopping condition for convergence

Gao et al. Efficient Knowledge Graph Accuracy Evaluation. PVLDB 2019

Iterative Procedure: Overview

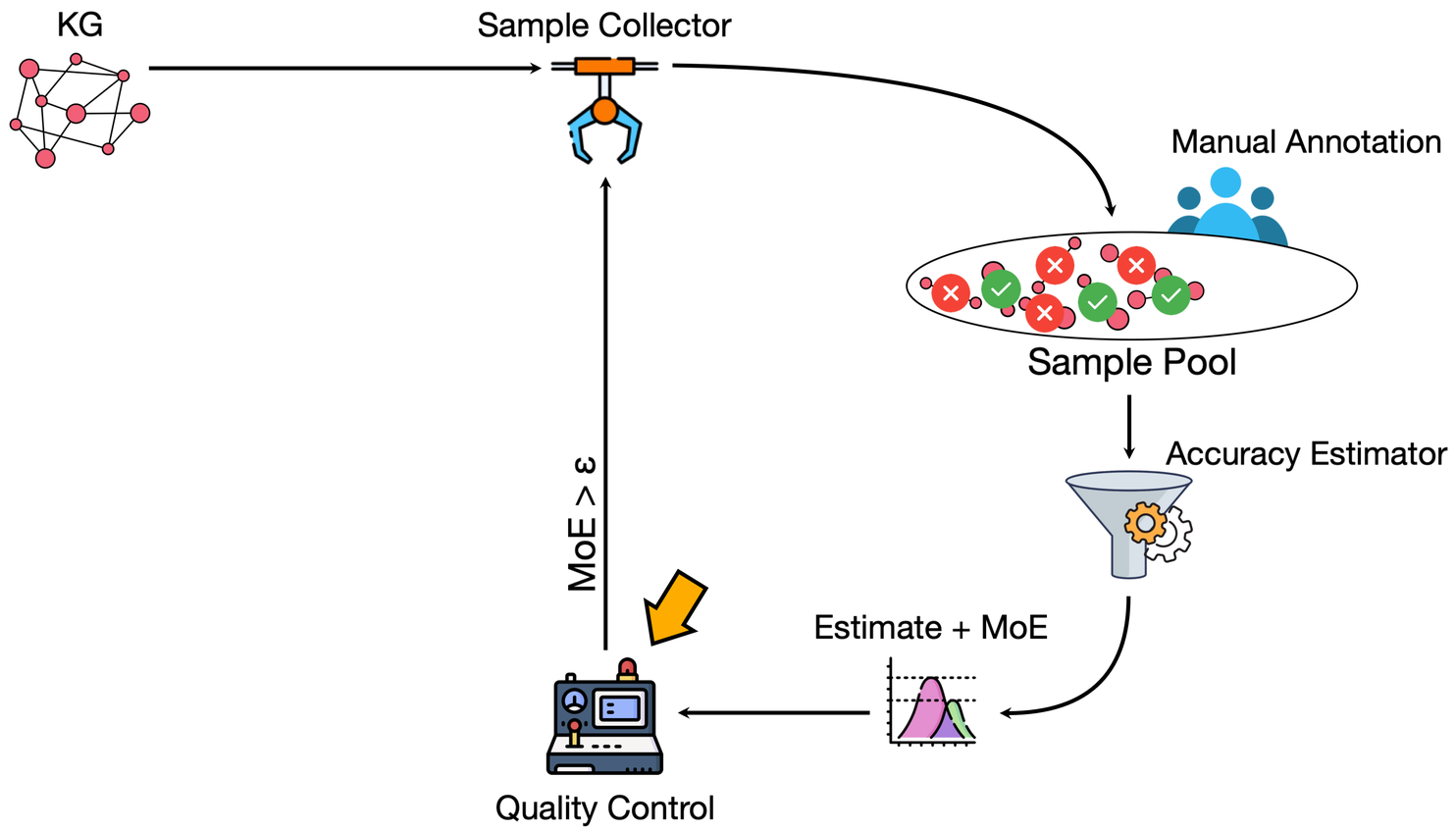
Sample, annotate, estimate, check the MoE



Gao et al. Efficient Knowledge Graph Accuracy Evaluation. PVLDB 2019

Iterative Procedure: Overview

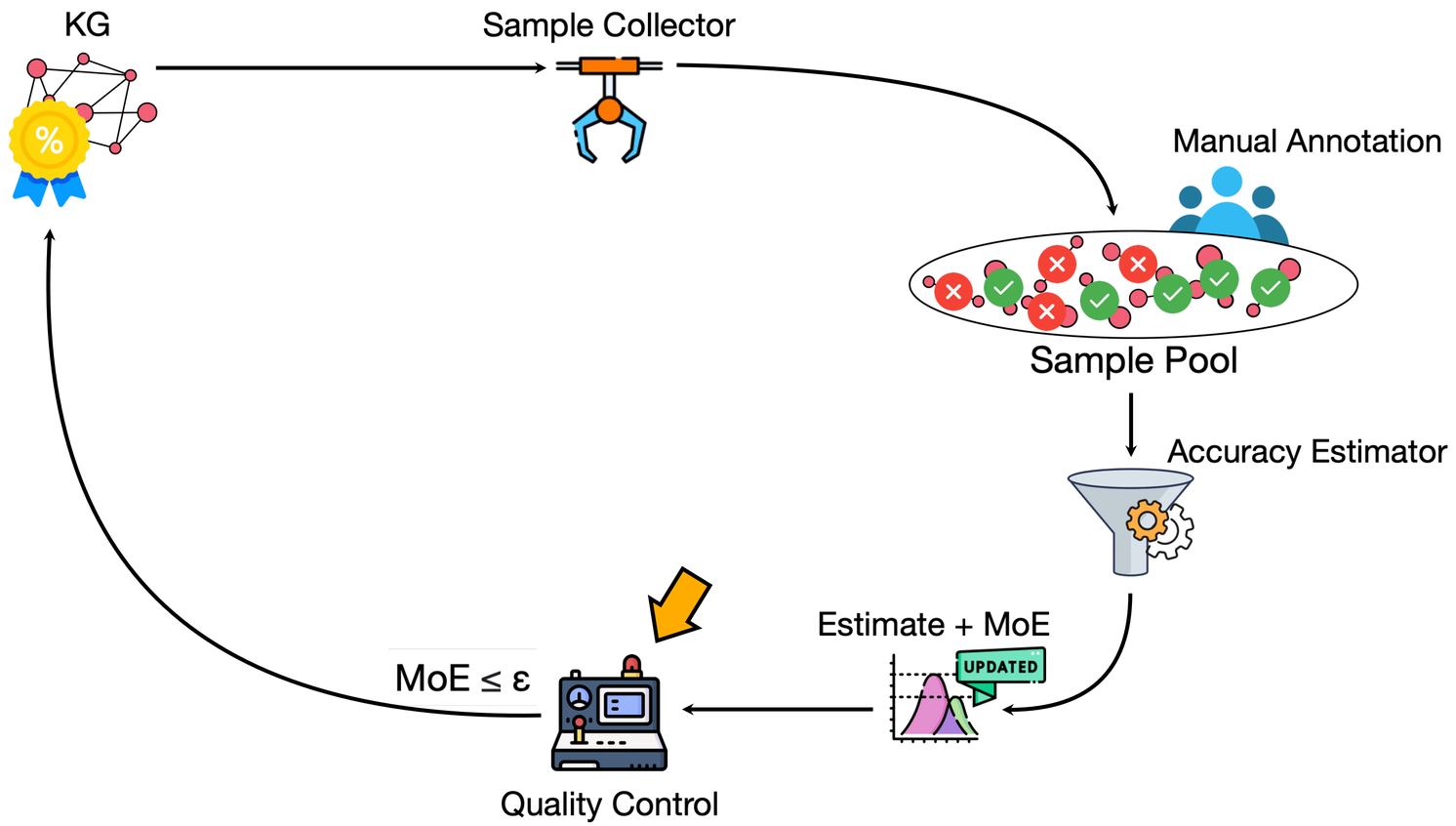
Sample, annotate, estimate, check the MoE



Gao et al. Efficient Knowledge Graph Accuracy Evaluation. PVLDB 2019

Iterative Procedure: Overview

Sample, annotate, estimate, check the MoE



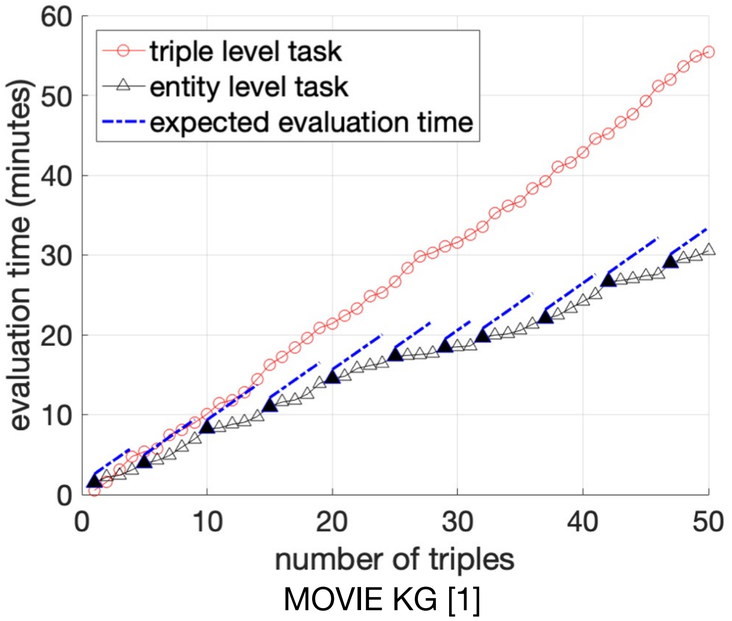
Gao et al. Efficient Knowledge Graph Accuracy Evaluation. PVLDB 2019

Annotation Cost: Evaluation Model

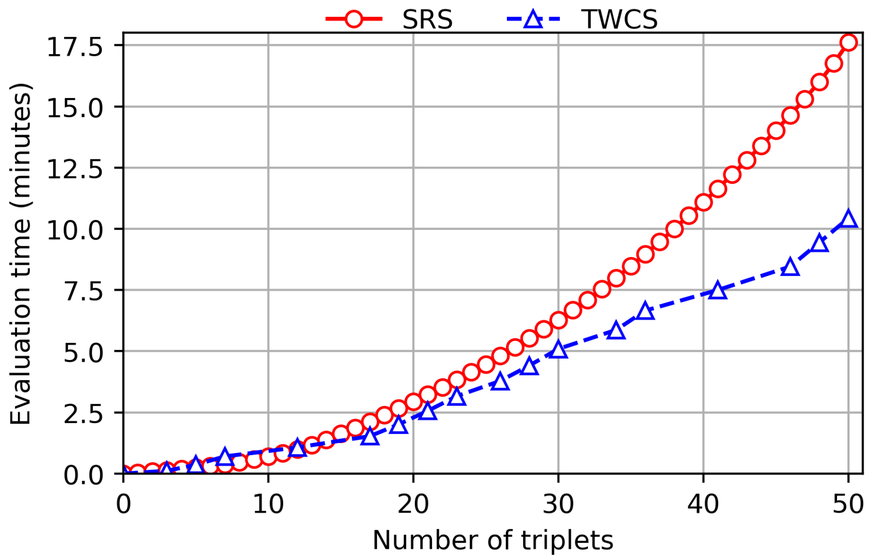
Identify the entity once, verify many facts

Annotating a new fact for an entity that has already been identified reduces the annotation cost compared to assessing a fact about an unseen entity.

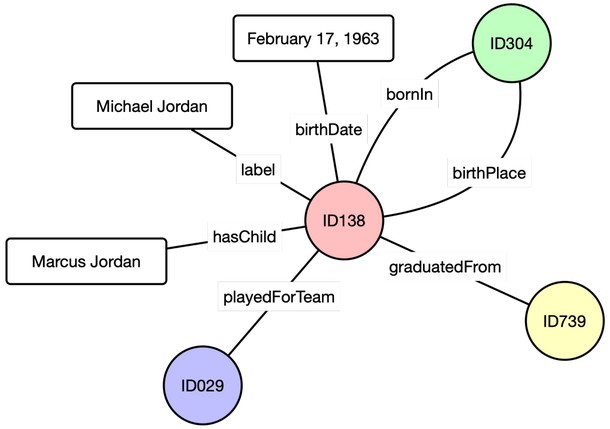
Entity clusters $G[e]$ become central aspects of the evaluation model.



MOVIE KG [1]



DBPEDIA KG [2]



[1] Gao et al. Efficient Knowledge Graph Accuracy Evaluation. PVLDB 2019 · [2] Marchesin et al. Utility-Oriented KG Accuracy Estimation with Limited Annotations. HCOMP 2024

Annotation Cost: Cluster-based Cost Function

Cost grows with entities and triples in the sample

Based on the evaluation model, the following cost function can be proposed

$$\text{cost}\big(\mathcal{S}(G)\big) = |E_{\mathcal{S}}| \cdot c_1 + |T_{\mathcal{S}}| \cdot c_2$$

where $\mathcal{S}(G)$ represents a sample from the KG, $E_{\mathcal{S}}$ the set of entities in the sample, $T_{\mathcal{S}}$ the set of triples in the sample, and c_1 and c_2 are the (estimated) average costs of auditing entities (**entity identification**) and triples (**fact verification**), respectively.

The cost function favors strategies capable of sampling subsets of triples sharing the same subject (i.e., entity clusters).

Gao et al. Efficient Knowledge Graph Accuracy Evaluation. PVLDB 2019

Three Sampling Designs

Sampling unit, estimator, variance [1]

Design	Sampling unit	Estimator	Variance
SRS	n triples, each drawn with probability $1/M$	$\hat{\mu}_s = \frac{1}{n}\sum_{i=1}^{n}\mathbf{1}(t_i)$	$V(\hat{\mu}_s) = \frac{\hat{\mu}_s(1-\hat{\mu}_s)}{n}$
TWCS	n clusters, drawn with probability proportional to size, with replacement [2]; then at most m triples per cluster by SRS	$\hat{\mu}_{w,m} = \frac{1}{n}\sum_{i=1}^{n}\hat{\mu}_i$	$V(\hat{\mu}_{w,m}) = \frac{1}{n(n-1)}\sum_{i=1}^{n}(\hat{\mu}_i - \hat{\mu}_{w,m})^2$
Stratified TWCS (STWCS)	Q strata of clusters, TWCS within each; stratum weight = its share of triples [3]	$\hat{\mu}_{ss} = \sum_{q=1}^{Q} W_q\,\hat{\mu}_q$	$V(\hat{\mu}_{ss}) = \sum_{q=1}^{Q} W_q^2\,V(\hat{\mu}_q)$

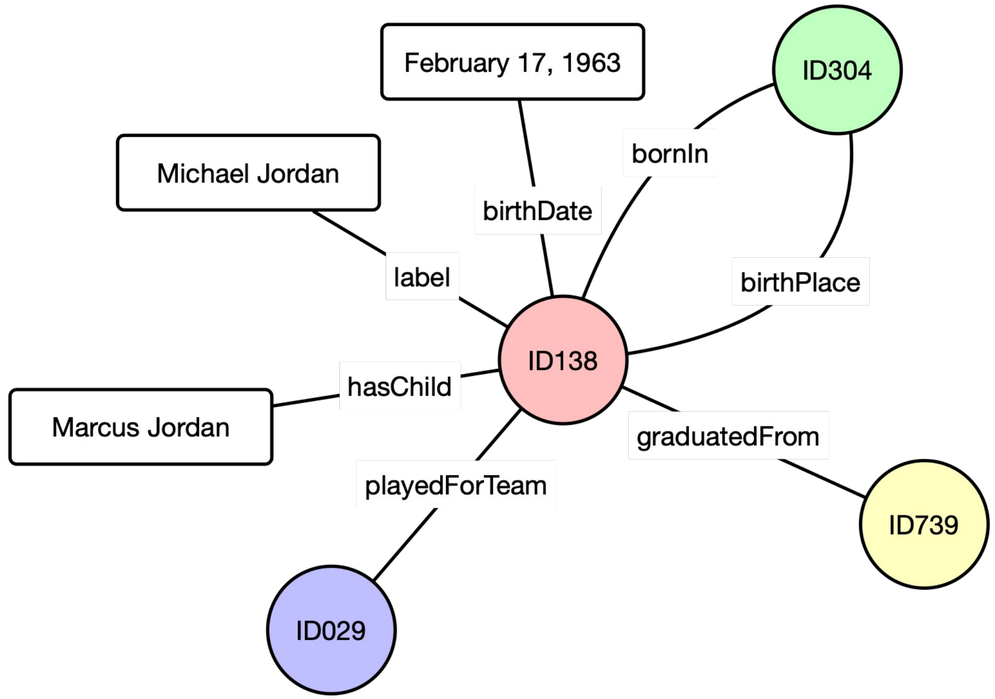
Estimators unbiased by design (Hansen-Hurwitz PPS; stratified weights); TWCS with m = 1 reduces to SRS [1, 4]

[1] Gao et al. Efficient Knowledge Graph Accuracy Evaluation. PVLDB 2019 · [2] Hansen and Hurwitz. Ann. Math. Statist. 1943
[3] Kish. Survey Sampling. 1965 · [4] Cochran. Sampling Techniques. 1977

Two-stage Weighted Cluster Sampling

The effectiveness of sampling entity clusters

TWCS: state-of-the-art design for KG accuracy estimation



TWCS consists of two stages:

Stage 1: sample n entity clusters with probability proportional to their sizes;

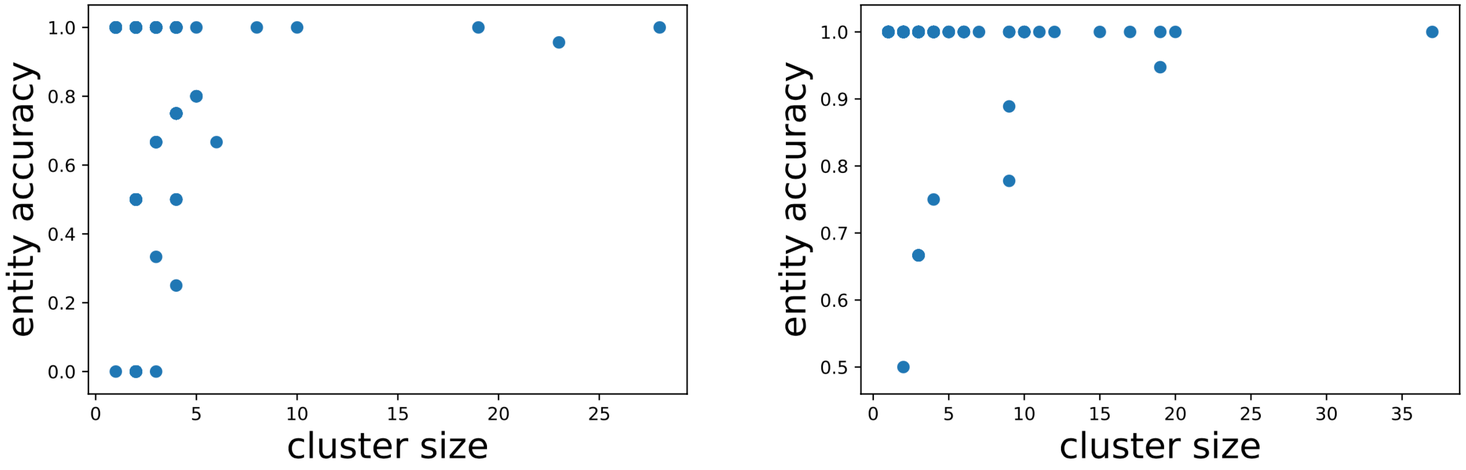
Stage 2: from each sampled cluster, sample at most m triples with SRS w/o replacement.

Second-stage size m between 3 and 5: near-minimum cost

Gao et al. Efficient Knowledge Graph Accuracy Evaluation. PVLDB 2019

Stratifying by Cluster Size

Cluster size as an accuracy proxy?



(a) NELL (b) YAGO

Why it helps: if the signal tracks accuracy, each stratum is homogeneous inside. The stratified estimator only pays for the variance within strata, so the same MoE needs fewer labels

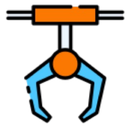
Gao et al. use cluster size: on NELL and YAGO, small clusters show the widest spread of accuracy

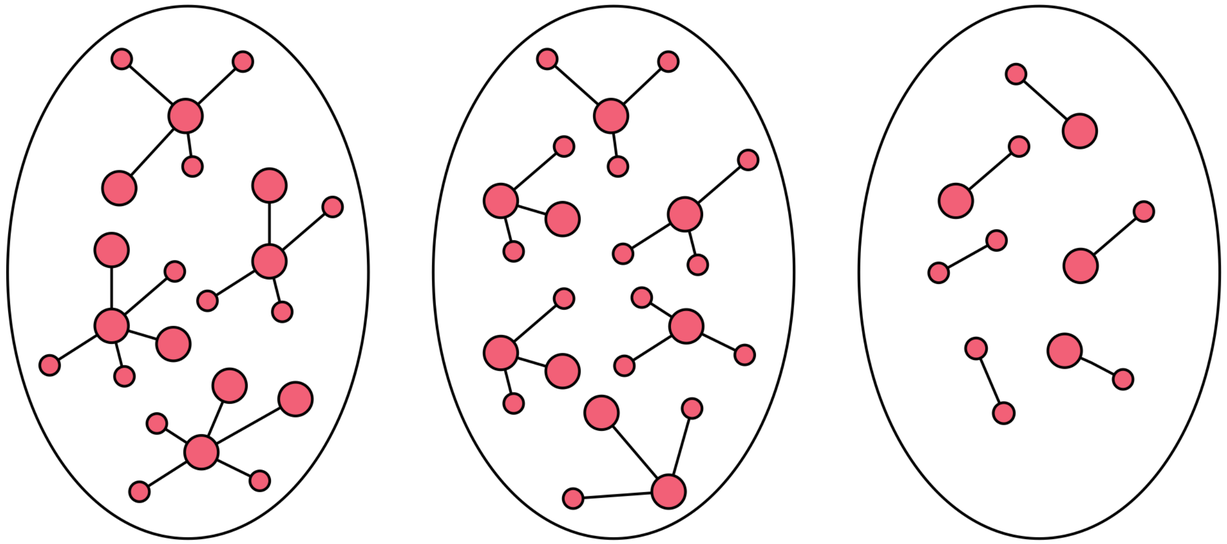
Question: is cluster size a good proxy for the accuracy of an entity?

Gao et al. Efficient Knowledge Graph Accuracy Evaluation. PVLDB 2019

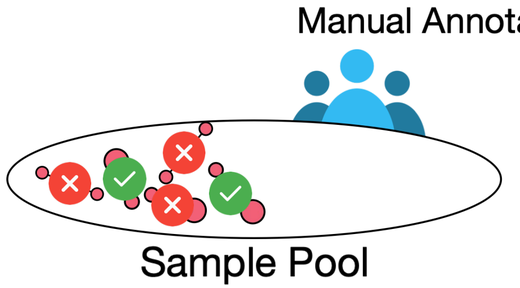
Stratified TWCS: Procedure

Pick a stratum by weight, run TWCS inside it





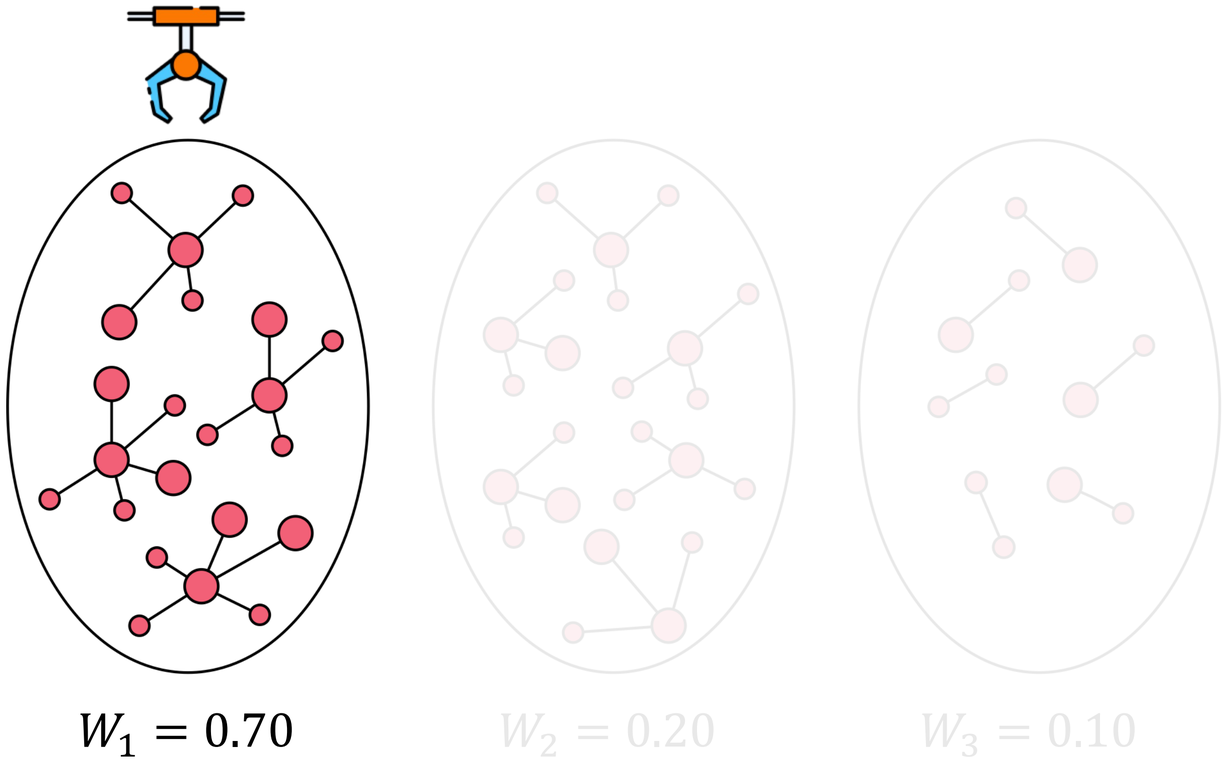
$W_1 = 0.70$ $W_2 = 0.20$ $W_3 = 0.10$



Gao et al. Efficient Knowledge Graph Accuracy Evaluation. PVLDB 2019

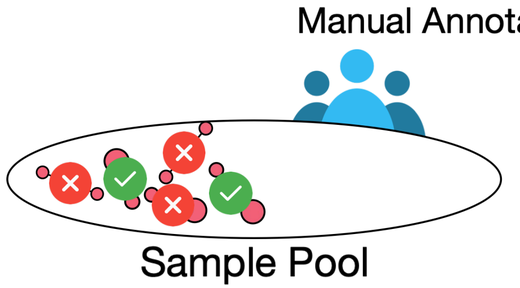
Stratified TWCS: Procedure

Pick a stratum by weight, run TWCS inside it



$W_1 = 0.70$ $W_2 = 0.20$ $W_3 = 0.10$

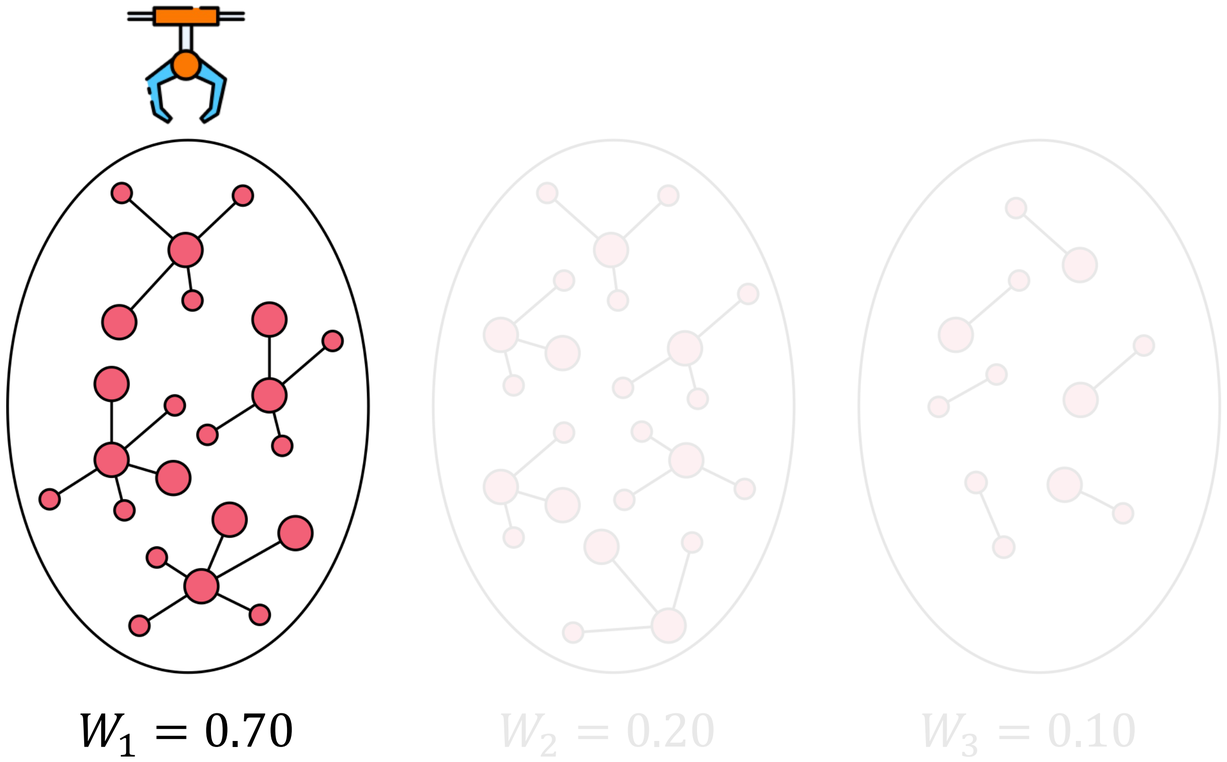
TWCS ($m = 2$)



Gao et al. Efficient Knowledge Graph Accuracy Evaluation. PVLDB 2019

Stratified TWCS: Procedure

Pick a stratum by weight, run TWCS inside it

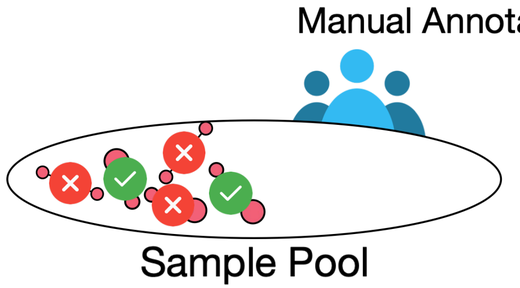


$W_1 = 0.70$ $W_2 = 0.20$ $W_3 = 0.10$

TWCS ($m = 2$)

Stage 1:

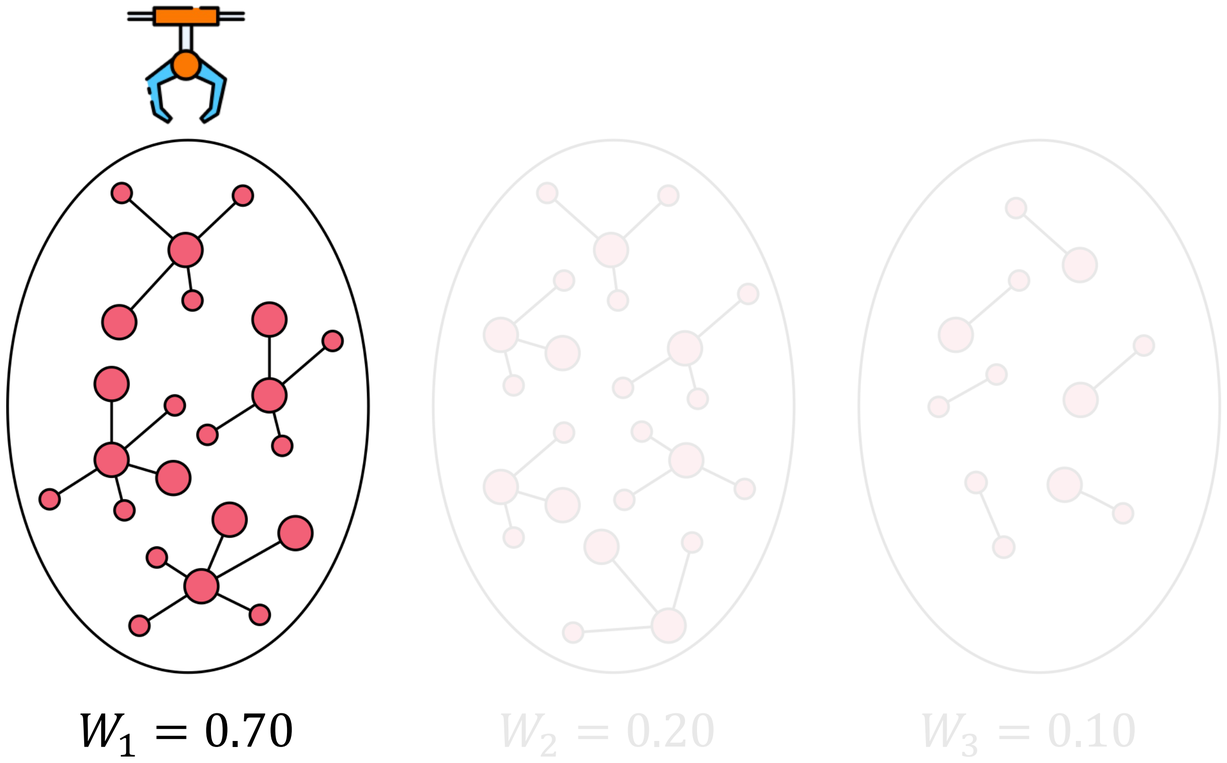
Stage 2:



Gao et al. Efficient Knowledge Graph Accuracy Evaluation. PVLDB 2019

Stratified TWCS: Procedure

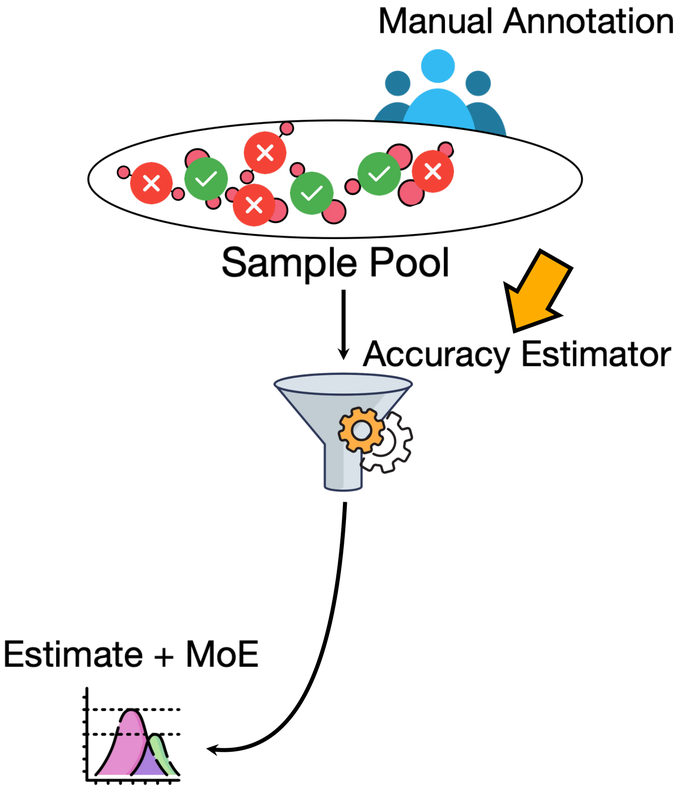
Pick a stratum by weight, run TWCS inside it



TWCS ($m = 2$)

Stage 1:

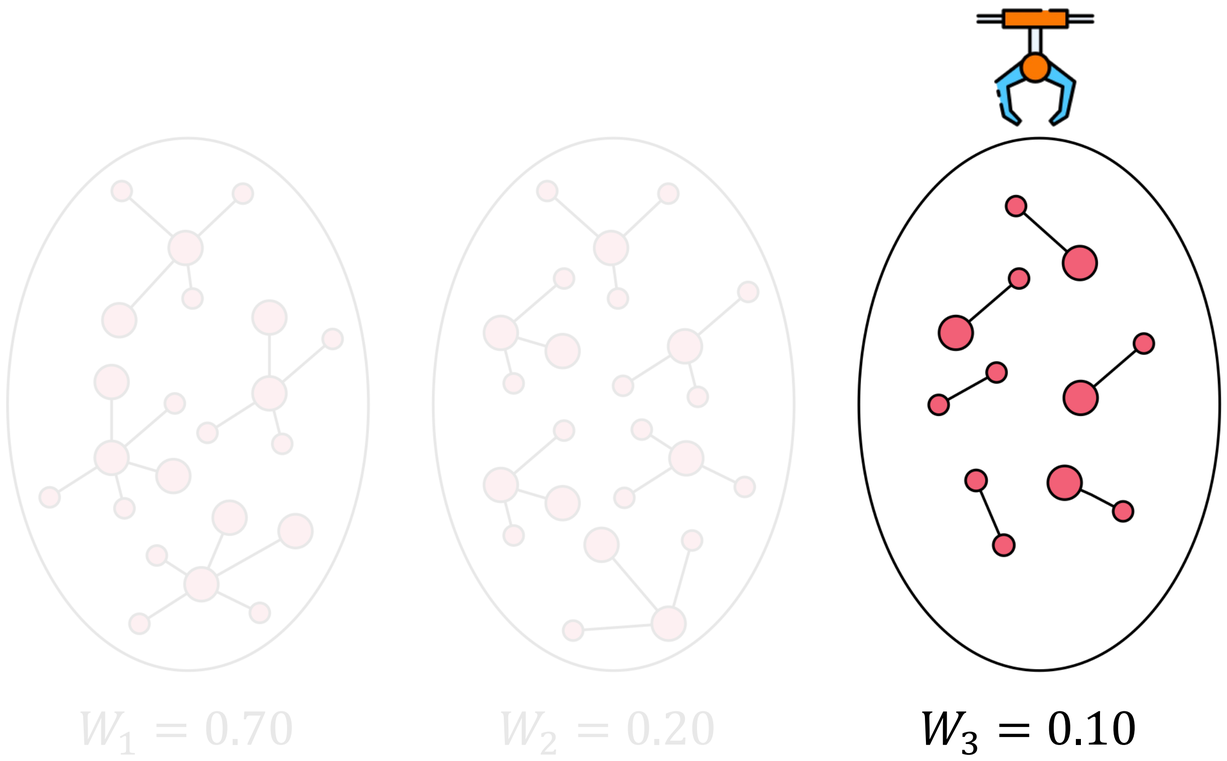
Stage 2:

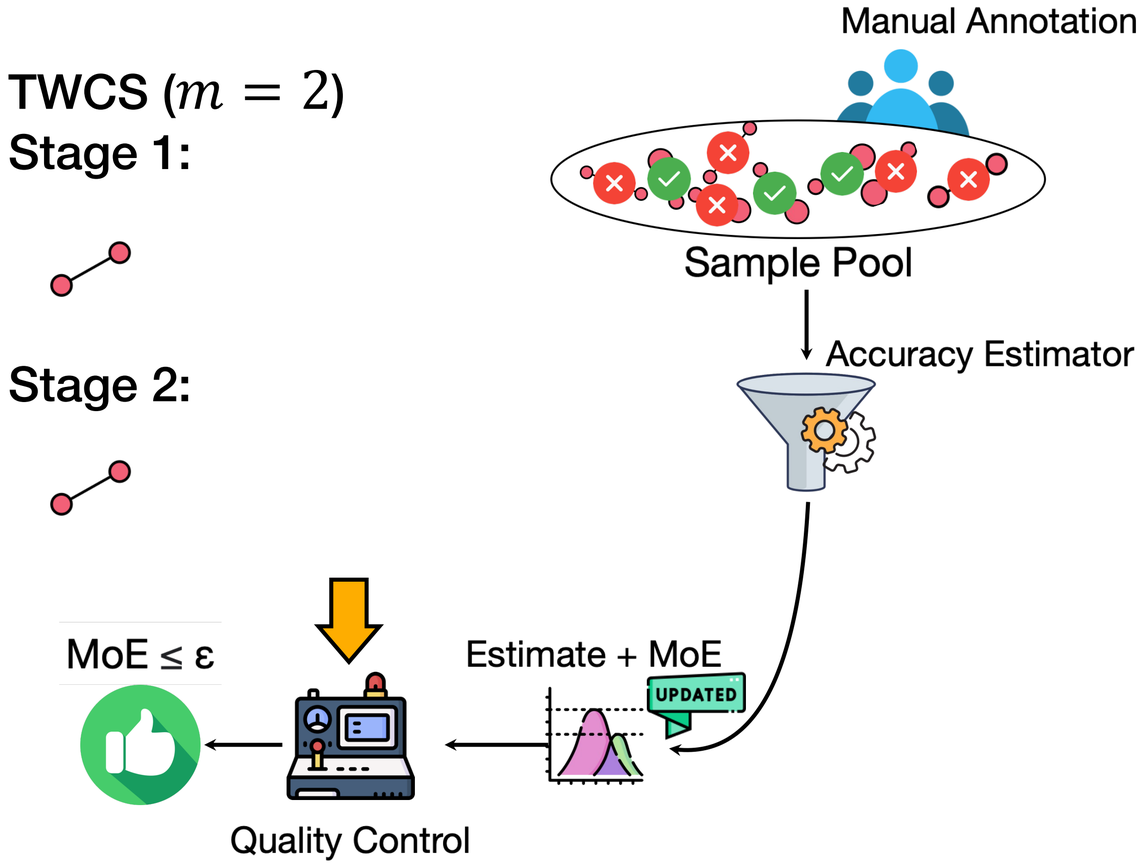


Gao et al. Efficient Knowledge Graph Accuracy Evaluation. PVLDB 2019

Stratified TWCS: Procedure

Pick a stratum by weight, run TWCS inside it





Gao et al. Efficient Knowledge Graph Accuracy Evaluation. PVLDB 2019

Take-Aways: 2.1

Sampling-based accuracy estimation



Define the estimand first: global, per predicate, per cluster



Match the design to the cost model: cluster sampling amortizes entity identification



Stratification pays only with an informative signal (see LLMs in 2.4)



Cluster sampling pays off: up to 60% less annotation time than SRS

Next: how far can these estimates be from the truth? (2.2)

Gao et al. Efficient Knowledge Graph Accuracy Evaluation. PVLDB 2019

Part 2: Evaluation Methodologies

2.1 Sampling-based accuracy estimation

2.2 From Wald to Wilson and Bayesian credible intervals

2.3 Utility-oriented and budget-aware estimation

2.4 LLMs for KG quality assessment

The Default: The Wald Interval

Inverting the large-sample normal test

Prior KG accuracy estimation relies on the Wald interval, obtained by **inverting the acceptance region** of the Wald large-sample normal test:

$$\left|\frac{(\hat{\mu}-\mu)}{\sqrt{V(\hat{\mu})}}\right| \leq z_{\alpha/2},$$

where $z_{\alpha/2}$ is the **critical value** of the standard normal distribution for a given significance level α.

Leading to the well-known formula: $\hat{\mu} \pm z_{\alpha/2} \cdot SE(\hat{\mu}) \equiv \sqrt{V(\hat{\mu})}$

Gao et al. Efficient Knowledge Graph Accuracy Evaluation. PVLDB 2019

Coverage Probability

Nominal vs empirical coverage

Robust CIs should contain the true parameter $1 - \alpha$ of the times over multiple repetitions of the sampling procedure.

Example: We repeat the sampling procedure 1000 times with a designated **confidence level** of 95% ($1 - \alpha$), we expect that in approximately 950 repetitions the true accuracy lies within the computed CI.

The confidence level is known as nominal coverage probability.
The value we get over the n repetitions is known as **empirical coverage probability.**

If empirical coverage is below nominal coverage, the CIs are said to be **permissive**.
The more permissive the CIs, the less reliable the sampling procedure.

Brown et al. Interval Estimation for a Binomial Proportion. Statistical Science 2001

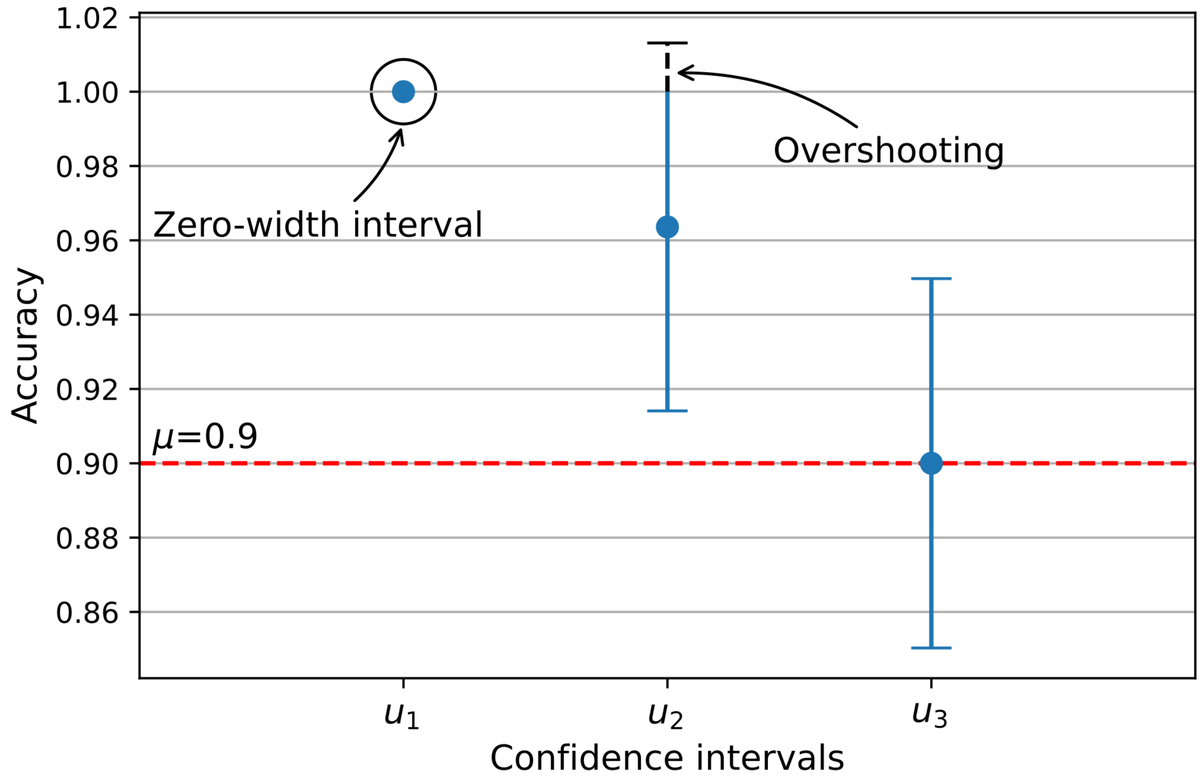
Limitations of the Wald Interval

Zero width, overshooting, poor coverage

Zero-width:
Implies absolute certainty.
Underestimation of the error.

Overshooting:
Exceeds [0, 1] boundaries.
Approximation fails.

Coverage:
Poor empirical coverage.
Reliability weakens [1, 2].

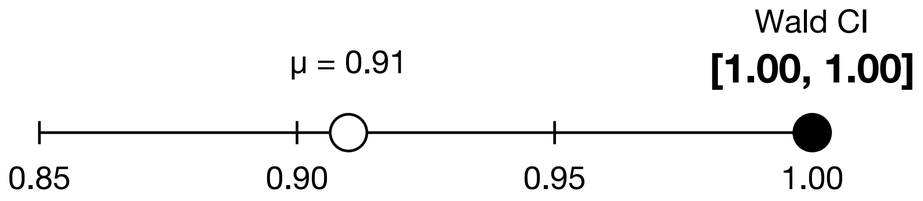


[1] Brown et al. Interval Estimation for a Binomial Proportion. Statist. Sci. 2001 · [2] Marchesin and Silvello. Efficient and Reliable Estimation of KG Accuracy. PVLDB 2024

A Zero-Width Interval in Practice

SRS and Wald on NELL (μ = 0.91), MoE threshold ε = 0.05

- Stops at $n = 30$ triples, all judged correct: $\hat{\mu} = 1.00$
- Estimated variance $\hat{\mu}(1 - \hat{\mu})/n = 0$, so the 95% Wald interval is $[1.00, 1.00]$
- Certainty from 30 labels, and the true value 0.91 lies outside
- This happened in 7% of 1,000 runs on NELL



Marchesin and Silvello. Efficient and Reliable Estimation of KG Accuracy. PVLDB 2024

The Wilson Interval

A score interval for binomial proportions [1]

Inverts the normal test with the **null standard error, not the estimated one:**

Relocated center estimate $\frac{\hat{\mu} + \frac{z^2_{\alpha/2}}{2n_S}}{1 + \frac{z^2_{\alpha/2}}{n_S}} \pm \frac{z_{\alpha/2}}{1 + \frac{z^2_{\alpha/2}}{n_S}} \cdot \sqrt{\frac{\hat{\mu}(1-\hat{\mu})}{n_S} + \frac{z^2_{\alpha/2}}{4n_S^2}}$ Corrected standard deviation

Wilson interval **solves zero-width** and **overshooting issues**
Provides **better coverage** than Wald interval

Caveat: Wilson assumes SRS
Requires design effect adjustments to be used with TWCS and STWCS [2, 3]

[1] Wilson. Probable Inference, the Law of Succession, and Statistical Inference. J. Amer. Statist. Assoc. 1927 · [2] Kish. Methods for Design Effects. J. Off. Stat. 1995 · [3] Marchesin and Silvello. Efficient and Reliable Estimation of KG Accuracy. PVLDB 2024

Wald vs Wilson on Real KGs

On YAGO, up to two times more reliable than Wald

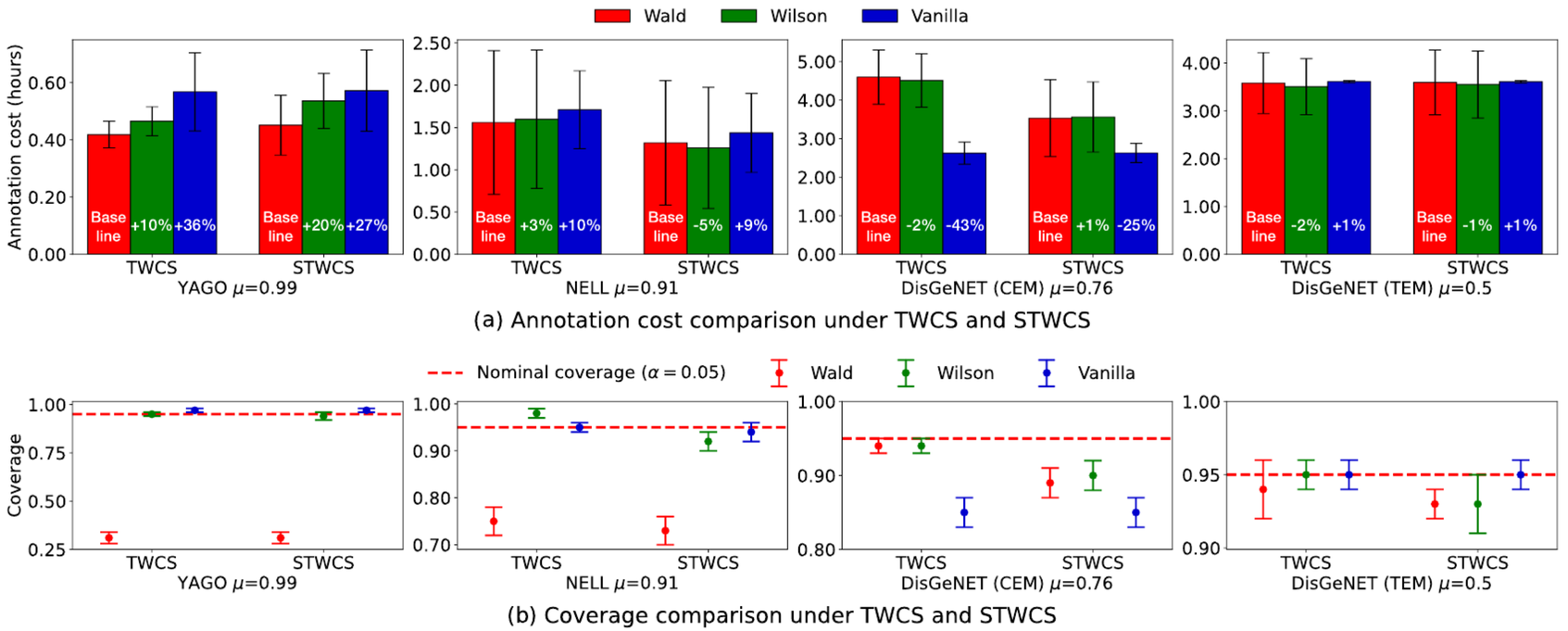


Figure: TWCS and STWCS; top, annotation cost relative to Wald; bottom, empirical coverage (nominal 0.95); Vanilla = Wilson without design effect



Wilson most **efficient** solution among **reliable** ones
Wilson best trade-off between efficiency and reliability

Marchesin and Silvello. Efficient and Reliable Estimation of KG Accuracy. PVLDB 2024

What a CI Does Not Tell Us

Three common misreadings [1, 2]

- **Fallacy 1:** "μ lies in this interval with probability 1 – α". After the data, a given CI contains μ or not; 1 – α describes the procedure
- **Fallacy 2:** "a narrow CI means a precise estimate". The zero-width Wald interval from 30 labels is not perfect knowledge [3]
- **Fallacy 3:** "the CI contains the likely values". A CI can exclude reasonable values; a zero-width one excludes all of them
- KG audits run once: practitioners need a one-shot interpretation

[1] Morey et al. The Fallacy of Placing Confidence in Confidence Intervals. Psychon. Bull. Rev. 2016
[2] Hoekstra et al. Robust Misinterpretation of Confidence Intervals. Psychon. Bull. Rev. 2014 · [3] Marchesin and Silvello. Credible Intervals for Knowledge Graph Accuracy Estimation. Proc. ACM Manag. Data (SIGMOD) 2025

Bayesian Credible Intervals

Post-data inference on KG accuracy

$$\mu \sim \mathrm{Beta}(a, b), \quad \tau_S \sim \mathrm{Bin}(n_S, \mu) \;\Rightarrow\; \mu \mid G_S \sim \mathrm{Beta}(a + \tau_S,\; b + n_S - \tau_S)$$

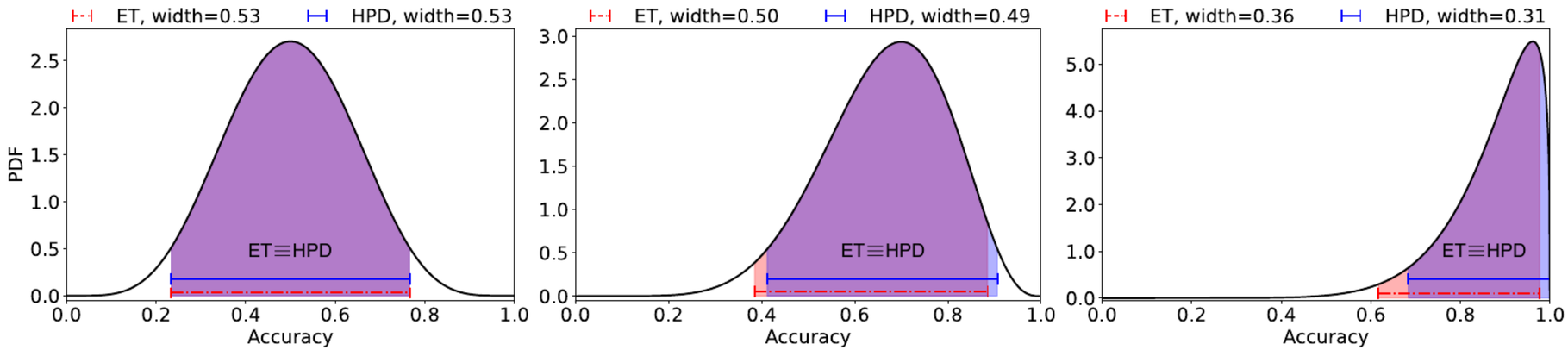
$$\Pr(l \leq \mu \leq u \mid G_S) = 1 - \alpha$$

- One-shot reading: μ lies in $[l, u]$ with probability 1 – α given the data [1–3]
- Conjugate update: add the correct and incorrect counts to a and b [1, 4]
- Width reflects the precision of the posterior

[1] Marchesin and Silvello. Credible Intervals for Knowledge Graph Accuracy Estimation. Proc. ACM Manag. Data (SIGMOD) 2025 · [2] Morey et al. Psychon. Bull. Rev. 2016
[3] Hoekstra et al. Psychon. Bull. Rev. 2014 · [4] Box and Tiao. Bayesian Inference in Statistical Analysis. 2011

Equal-Tailed vs HPD Intervals

Shortest interval for skewed posteriors

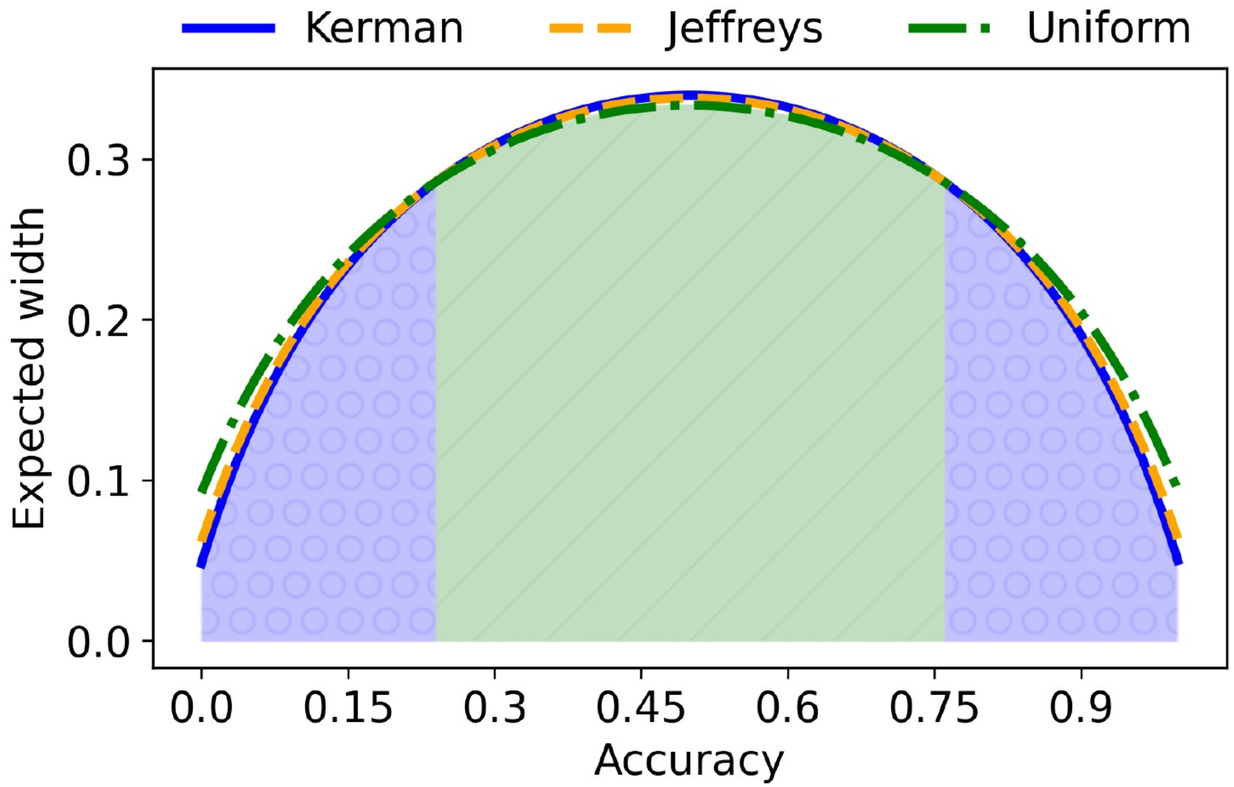


- Equal-tailed (ET): $\alpha/2$ in each tail; wastes width when the posterior is skewed
- HPD: every point inside has higher density than any point outside
- HPD is the shortest $1 - \alpha$ interval and is unique; it equals ET for symmetric posteriors
- KG accuracy is often far from 0.5 (towards 1): posteriors are skewed

Marchesin and Silvello. Credible Intervals for KG Accuracy Estimation. Proc. ACM Manag. Data (SIGMOD) 2025

Choosing the Prior

No uninformative prior is shortest everywhere



Expected HPD width for $n = 30$ labels, $\alpha = 0.05$

- Kerman Beta(1/3, 1/3): shortest near 0 and 1 [1, 2]
- Uniform Beta(1, 1): shortest near 0.5 [1]
- Jeffreys Beta(1/2, 1/2): never the shortest of the three [1, 3]
- Where $\hat{\mu}$ will fall is unknown before annotating

[1] Marchesin and Silvello. Credible Intervals for Knowledge Graph Accuracy Estimation. Proc. ACM Manag. Data (SIGMOD) 2025 · [2] Kerman. Neutral Noninformative and Informative Conjugate Beta and Gamma Prior Distributions. Electron. J. Stat. 2011
[3] Jeffreys. An Invariant Form for the Prior Probability in Estimation Problems. Proc. R. Soc. A 1946

αHPD: adaptive HPD

Let the priors compete

1. Sample and annotate a batch (SRS or TWCS); adjust the counts for the design effect [1, 2]
2. Update one Beta posterior per uninformative prior: Kerman, Jeffreys, Uniform [1]
3. HPD interval per posterior: closed form if all labels agree, SLSQP optimization otherwise [1, 3]
4. Keep the shortest interval; stop when its MoE ≤ ε, otherwise return to step 1

Informative priors plug in unchanged when reliable prior knowledge exists (2.3)

[1] Marchesin and Silvello. Credible Intervals for Knowledge Graph Accuracy Estimation. Proc. ACM Manag. Data (SIGMOD) 2025 · [2] Kish. Methods for Design Effects. J. Off. Stat. 1995
[3] Kraft. A Software Package for Sequential Quadratic Programming. DLR Tech. Rep. 1988

αHPD vs Wald and Wilson

Annotation cost in hours, TWCS with m = 3

KG (μ)	Wald	Wilson	aHPD
YAGO (μ = 0.99)	0.42 ± 0.05	0.47 ± 0.06	**0.41 ± 0.03**
NELL (μ = 0.91)	1.58 ± 0.87	1.62 ± 0.84	**1.40 ± 0.85**
DBpedia (μ = 0.85)	2.80 ± 0.87	2.69 ± 0.84	**2.55 ± 0.95**
FactBench (μ = 0.54)	**3.08 ± 0.45**	3.11 ± 0.48	3.11 ± 0.48

DBpedia: gold subset of 9,344 facts (μ = 0.85); the DBpedia study of 2.3 estimates 0.83 on 9,930 facts

- αHPD is significantly cheaper than both on the skewed KGs (t-test, $p < 0.01$) and ties Wilson on FactBench
- At $\alpha = 0.01$ the saving over Wilson reaches 47% (SRS) and 39% (TWCS) on YAGO

Marchesin and Silvello. Credible Intervals for KG Accuracy Estimation. Proc. ACM Manag. Data (SIGMOD) 2025

Take-Aways: 2.2

Interval estimation

Interval	Failure modes	Interpretation	Cost
Wald	zero width, overshoot	repeated-sampling	cheap but unreliable
Wilson	neither	repeated-sampling	reliable, costlier near 0 and 1
aHPD	neither	one-shot posterior	reliable; cheapest on skewed KGs



Report the interval method with every KG accuracy figure

Next: should every fact count the same? (2.3)

Marchesin and Silvello. Efficient and Reliable Estimation of KG Accuracy. PVLDB 2024 · Marchesin and Silvello. Credible Intervals for Knowledge Graph Accuracy Estimation. Proc. ACM Manag. Data (SIGMOD) 2025

Part 2: Evaluation Methodologies

2.1 Sampling-based accuracy estimation
2.2 From Wald to Wilson and Bayesian credible intervals
2.3 Utility-oriented and budget-aware estimation
2.4 LLMs for KG quality assessment

Utility: Are All Facts Equal?

Motivation for utility-oriented evaluation

Different parts of the KG may have **varying utility** depending on the downstream task

- For entity-oriented search, **popular entities** carry the **highest query load**
  - **Errors there are seen by more queries**

Utility can also help when dealing with **limited annotation budgets**

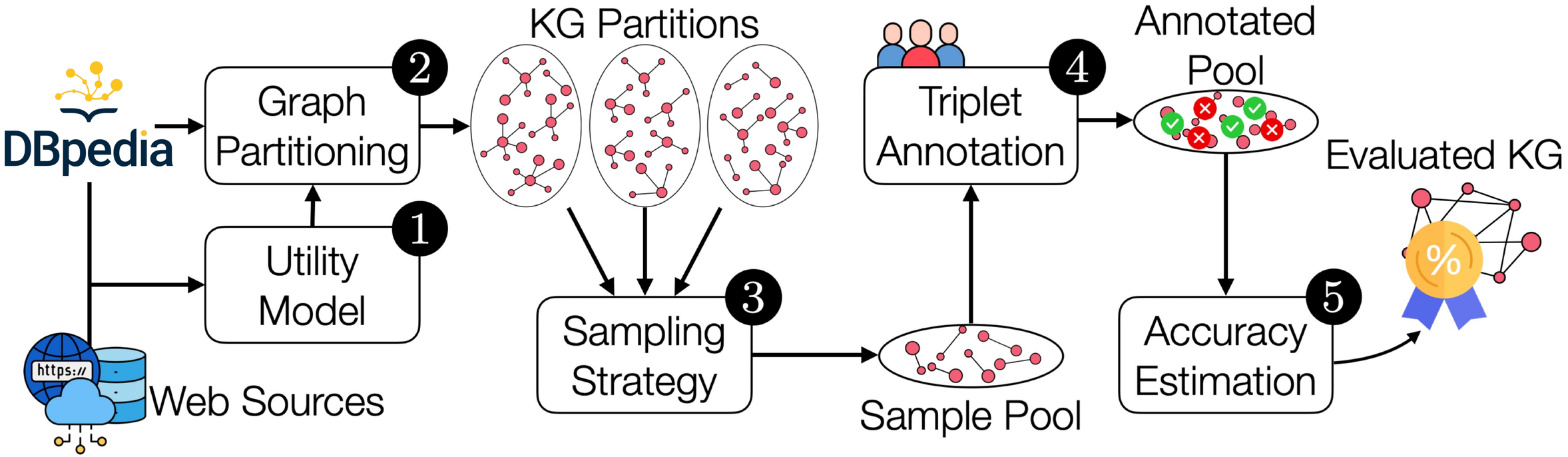
- Can guide where the annotation budget goes
- Helps decide when to apply **filtering** or **correction mechanisms** in low-quality cases

Marchesin et al. Utility-Oriented KG Accuracy Estimation with Limited Annotations: A Case Study on DBpedia. HCOMP 2024

Utility-Oriented Evaluation Framework

DBpedia 2015-10 case study

6.2M entities and 1.1B triples; 170M A-Box triples after filtering



Marchesin et al. Utility-Oriented KG Accuracy Estimation with Limited Annotations: A Case Study on DBpedia. HCOMP 2024

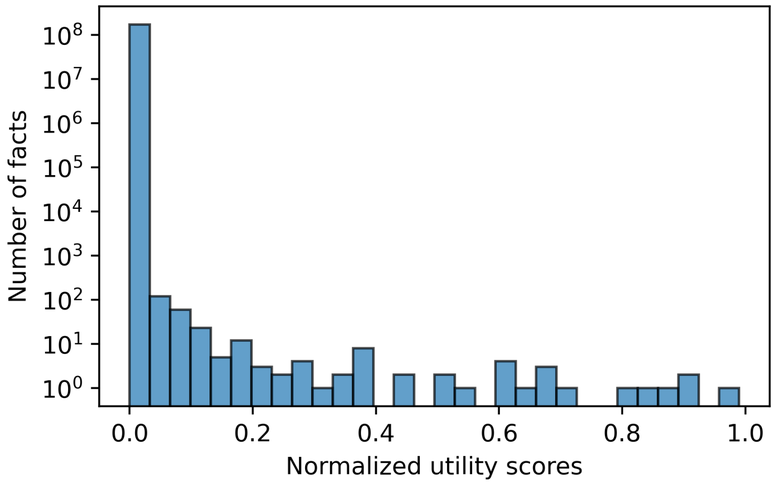
Utility from SPARQL Query Logs

Popularity of facts in SPARQL query logs

Utility of a fact = how often it appears in query results or in their lineage (queries rewritten as CONSTRUCT) [1]

The DBpedia study uses the **LSQ 2.0 logs [2]** for DBpedia 2015-10:

- **1.67M SELECT** queries (98%)
- 31K CONSTRUCT, 25 ASK, and 6 DESCRIBE queries (2%)





Most facts cluster near zero, but there is a notable **long tail of high utility facts**

[1] Dosso et al. Credit Distribution in Relational Scientific Databases. Inf. Syst. 2022
[2] Stadler et al. LSQ 2.0: A Linked Dataset of SPARQL Query Logs. Semantic Web 2024

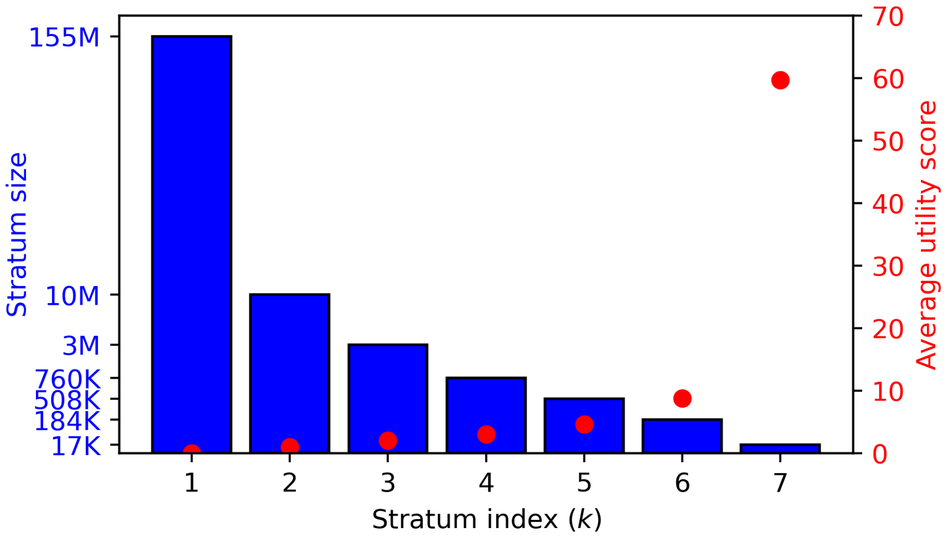
Stratifying Facts by Utility

Seven strata from the utility distribution

Facts are partitioned into **strata** by utility score

Stratification is done using **Cumulative Square Root of Frequency (CSRF)**

- CSRF minimizes **intra-stratum variance** in utility scores





The number of partitions ($k = 7$) strikes a **balance** between **capturing diverse aspects** and **keeping a reasonable granularity**



Large stratum with **near-zero utility scores** and several **smaller** strata with **higher utility scores**

Dalenius and Hodges. Minimum Variance Stratification. JASA 1959

Allocating Budget Across Strata

Per-stratum precision and budget targets

$$\min_{S} \ \mathrm{cost}(S(P_i)) \quad \text{s. t.} \quad E[\hat{\mu}_i] = \mu(P_i) \ \wedge \ (\mathrm{MoE}(\hat{\mu}_i, \alpha) \leq \varepsilon_i \ \vee \ |S(P_i)| = b_i)$$

- **Formulation:** each stratum gets its own precision target ε and budget b; its sampling stops at whichever comes first [1]
- **Design:** tighter ε or larger b for high-utility strata, looser targets for low-utility ones
- The KG-level estimate stays unbiased: strata are combined with their design weights (share of facts) [2, 3]
- **Reported so far:** no study has varied the targets by utility. CIKM 2024 used the same ε = 0.05 in every stratum; HCOMP 2024 annotated 500 TWCS batches in every stratum [2]

[1] Marchesin et al. Veracity Estimation for Entity-Oriented Search with Knowledge Graphs. CIKM 2024
[2] Marchesin et al. Utility-Oriented KG Accuracy Estimation with Limited Annotations. HCOMP 2024 · [3] Cochran. Sampling Techniques. 1977

DBpedia: Partition and KG Estimates

Seven utility strata, 95% intervals

37,546 annotations on 11,296 facts from 60 annotators; 9,930 facts retained

	Correct	Incorrect	IDK	Total	Estimate
P1	1,851	376	200	2,427	0.83±0.02
P2	1,416	281	110	1,807	0.85±0.02
P3	1,343	246	105	1,694	0.85±0.02
P4	1,164	164	60	1,388	0.89±0.02
P5	927	133	74	1,134	0.89±0.02
P6	715	99	22	836	0.90±0.03
P7	533	96	15	644	0.87±0.03
KG	7,949	1,395	586	9,930	0.83±0.02

IDK = I don't know; IDK labels are excluded from the estimates



DBpedia 2015-10 is of **high quality**

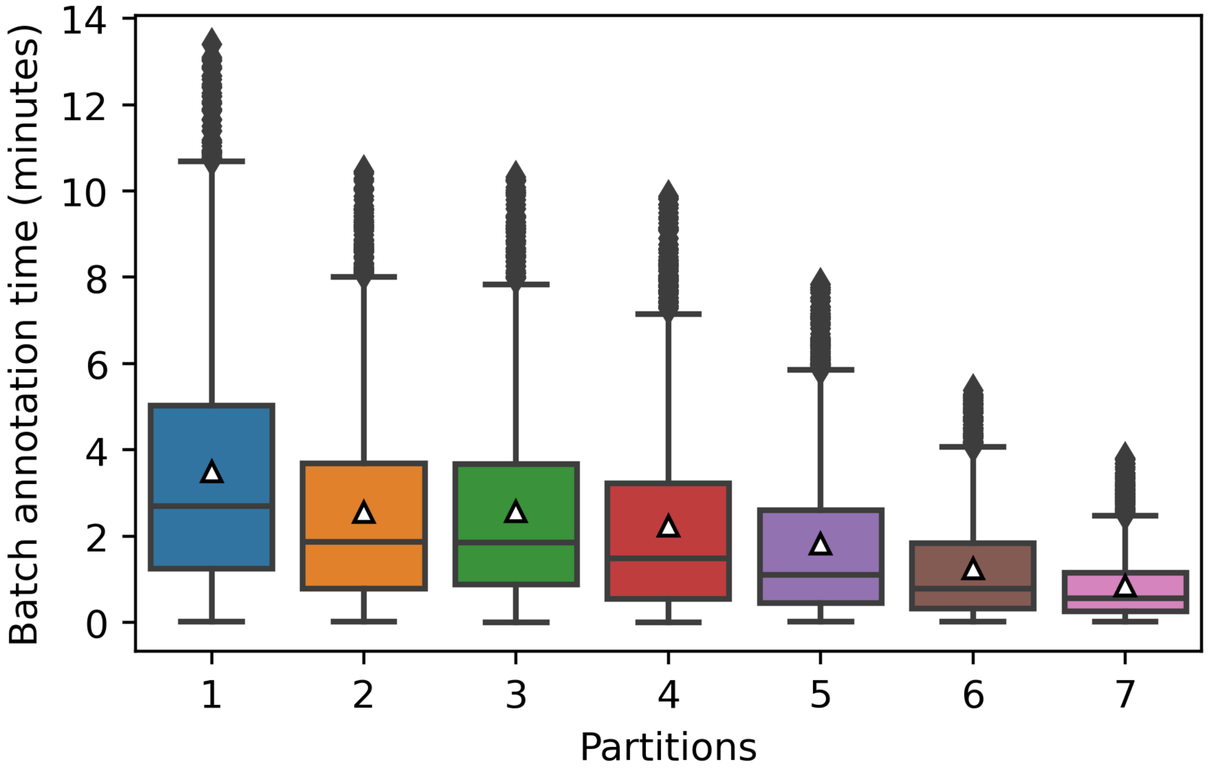


95% intervals; with over 600 labeled facts per stratum and estimates of 0.83 to 0.90

Marchesin et al. Utility-Oriented KG Accuracy Estimation with Limited Annotations: A Case Study on DBpedia. HCOMP 2024

Annotation Time per Partition

Batch annotation time per partition





Annotating batches from **low-utility** partitions demands **more time** than high-utility ones (P1 needs over twice the time of P7)



Costs differ by stratum, so an allocation should account for them

Marchesin et al. Utility-Oriented KG Accuracy Estimation with Limited Annotations: A Case Study on DBpedia. HCOMP 2024

Heavy Tails and Cold Start

Practical issues of utility-oriented designs

- Heavy tail: one near-zero-utility stratum holds most facts and drives the KG-level estimate (DBpedia: 0.83) [1]
- **Bias control (design):** if popular strata get more budget, keep the design weights in the estimator [2]
- Popularity is not an accuracy proxy: veracity and popularity appear nearly orthogonal [3]
- Cold start without query logs: web popularity [3, 4], cluster size [5], or LLM signals (2.4)
- **Design:** start with proportional allocation; shift budget as stratum estimates accrue

[1] Marchesin et al. Utility-Oriented KG Accuracy Estimation with Limited Annotations. HCOMP 2024 · [2] Cochran. Sampling Techniques. 1977 · [3] Marchesin et al. Veracity Estimation for Entity-Oriented Search with Knowledge Graphs. CIKM 2024
[4] Zheng et al. Crowdsourced Fact Validation for Knowledge Bases. ICDE 2022 · [5] Gao et al. Efficient Knowledge Graph Accuracy Evaluation. PVLDB 2019

Take-Aways: 2.3

Utility-oriented estimation



Utility says where errors hurt; accuracy says where errors are: account for both



Design: allocate budget by utility, weight estimates by design



Reported studies used even allocation and frequentist intervals



Without query logs, utility needs a proxy: web popularity, cluster size, LLMs

Next: can LLMs provide the signal, or even the labels? (2.4)

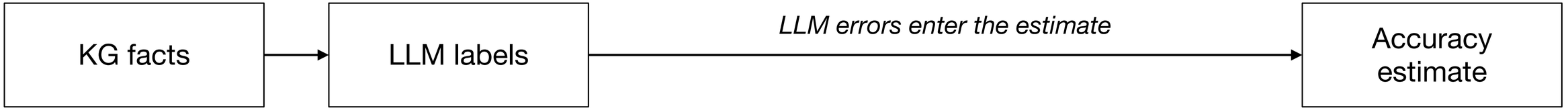
Part 2: Evaluation Methodologies

2.1 Sampling-based accuracy estimation
2.2 From Wald to Wilson and Bayesian credible intervals
2.3 Utility-oriented and budget-aware estimation
2.4 LLMs for KG quality assessment

Two Roles for LLMs

Validator or signal

Validator



Signal



LLM errors only cost efficiency: the estimate uses human labels

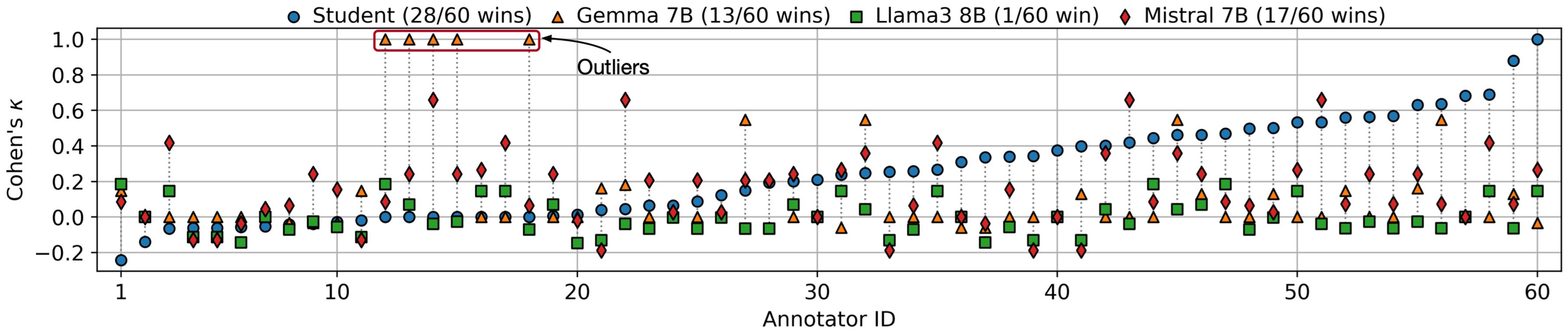
Question 1: can LLMs replace annotators? [1]

Question 2: can LLMs make human annotation cheaper? [2]

[1] Shami et al. Benchmarking Large Language Models for Knowledge Graph Validation. EDBT 2026
[2] Marchesin et al. LLMs as Stratification Signals for KG Accuracy Evaluation. PVLDB 2026

Pilot: LLMs vs Student Annotators

Honey-pot agreement and annotation time



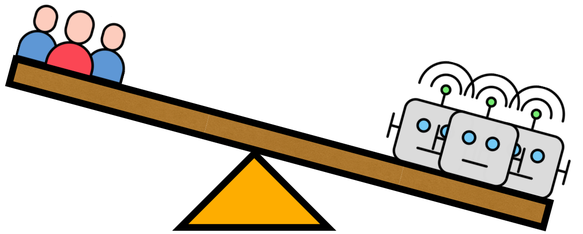
Reliability:

- LLMs outperform 53% of the students in honey-pot evaluations

Efficiency:

- LLM average annotation time: 3.48 hours for the entire set of facts
- Student average annotation time: 5.21 days for the same task

Good! … Right?



Marchesin et al. Utility-Oriented KG Accuracy Estimation with Limited Annotations: A Case Study on DBpedia. HCOMP 2024

Pilot: Against Expert Labels

All expert labels, three open LLMs

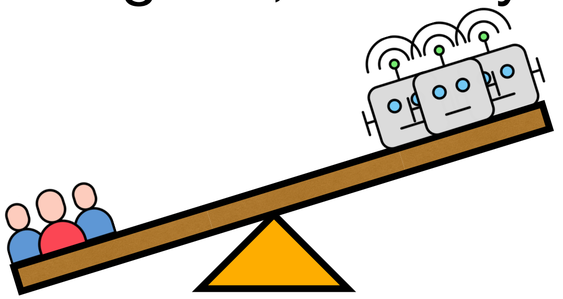
	Gemma 7B	Llama3 8B	Mistral 7B
Compliance	1.00	0.89	1.00
Truthfulness	0.78	0.82	0.90
Informativeness	1.00	0.82	0.66
Accuracy	0.78	0.65	0.59
Balanced Accuracy	0.36	0.35	0.46

Compliance: ability to follow instructions

Truthfulness: ability to provide honest responses (correct answer or IDK)

Informativeness: ability to provide informative responses (everything but IDK)

Not good, actually!





Low balanced accuracy due to **LLMs' reluctance** to provide **incorrect** responses

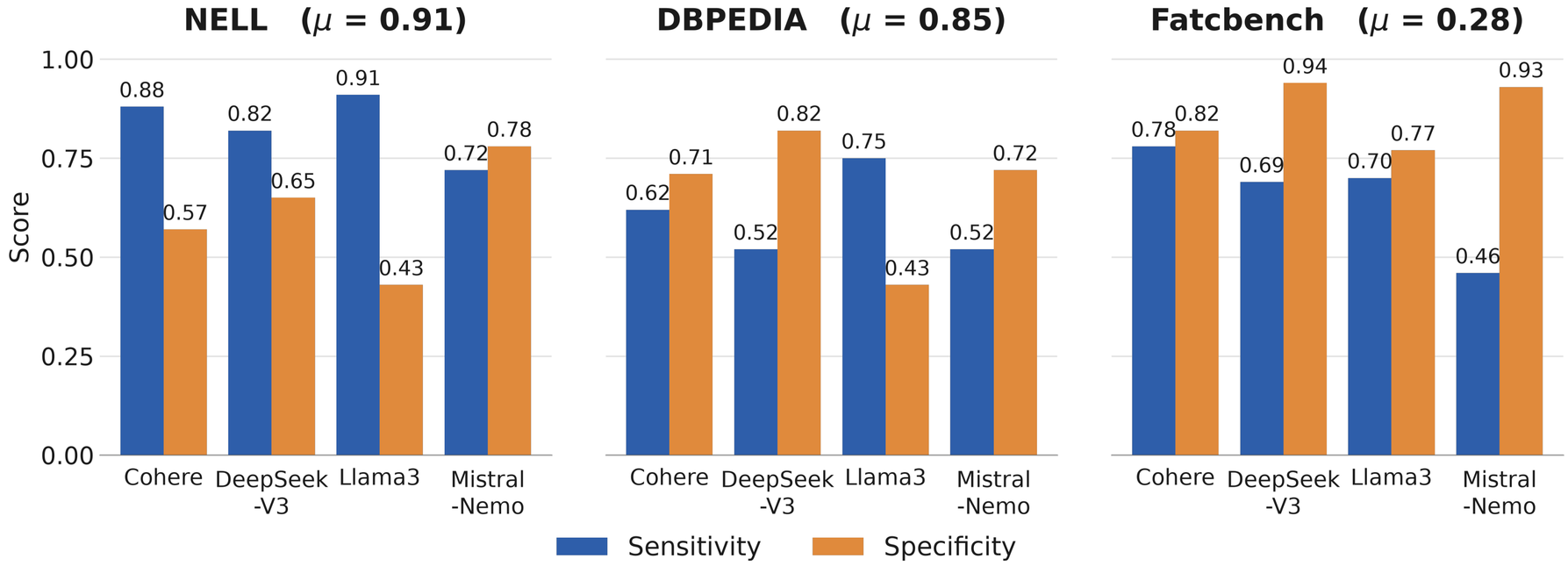


Lack of incorrect responses led to **overestimated accuracy scores** approaching 100%

Marchesin et al. Utility-Oriented KG Accuracy Estimation with Limited Annotations: A Case Study on DBpedia. HCOMP 2024

LLMs as Direct KG Evaluators

Dataset-dependent, asymmetric errors



high sensitivity | **high variability** | **high specificity**

LLM-derived KG accuracy deviates up to 45% from ground truth

Marchesin et al. LLMs as Stratification Signals for KG Accuracy Evaluation. PVLDB 2026

FactCheck: Benchmarking LLM Validators

Can LLMs validate KG facts? A systematic test

The pilots of HCOMP 2024 and PVLDB 2026 used few models and one prompting setup. FactCheck tests the validator role along three questions:

RQ1 · Internal knowledge

Can LLMs validate facts using only their parametric knowledge?

RQ2 · External evidence (RAG)

Does evidence retrieved from the web improve validation?

RQ3 · Multi-model consensus

Does aggregating several LLMs make validation more reliable?

3

KG datasets: FactBench, YAGO, DBpedia

2M+

web documents retrieved as evidence

5

LLMs: four open models and GPT-4o mini

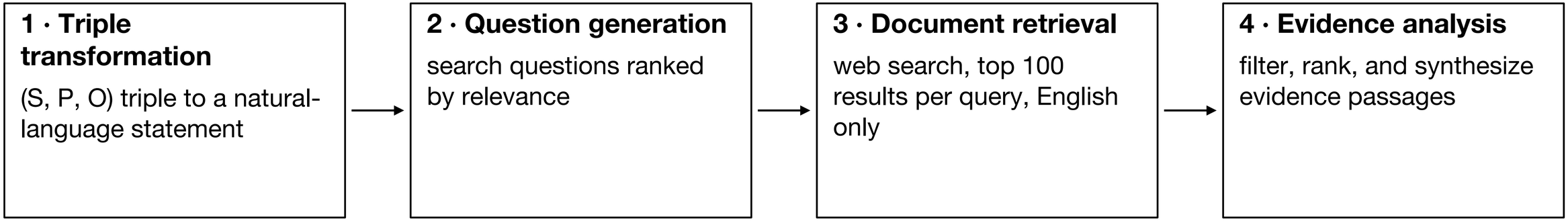
4

validation strategies: DKA, GIV-Z, GIV-F, RAG

Shami et al. Benchmarking Large Language Models for Knowledge Graph Validation. EDBT 2026

RAG for Fact Validation

Retrieval pipeline, gains, and cost



Cost: RAG takes 1.6–2.9 s per triple vs 0.17–0.37 s for direct prompting (DKA): often six or more times slower

Gains: RAG can outperform internal knowledge, but gains are inconsistent across datasets

Bias: on DBpedia, RAG often underperforms: retrieved evidence can introduce contextual bias

Trade-off: higher compute cost does not buy proportionally higher accuracy

Shami et al. Benchmarking Large Language Models for Knowledge Graph Validation. EDBT 2026

FactCheck: Best F1 per Setting

Best model per strategy; F1 on true / false facts

Strategy	FactBench (μ = 0.54)	YAGO (μ = 0.99)	DBpedia (μ = 0.85)
DKA (internal knowledge)	Gemma 2 · 0.75 / 0.74	Gemma 2 · 0.82 / 0.02	Gemma 2 · 0.85 / 0.36
GIV-Z (zero-shot)	Mistral · 0.77 / 0.72	Gemma 2 · 0.88 / 0.03	Mistral, Gemma 2 · 0.87 / 0.37
GIV-F (few-shot)	Mistral, Gemma 2 · 0.81 / 0.76	Gemma 2, Qwen 2.5 · 0.92 / 0.03	Mistral, Qwen 2.5 · 0.89 / 0.36
RAG	GPT-4o mini · 0.91 / 0.90	Mistral, Gemma 2 · 0.96 / 0.03	Qwen 2.5 · 0.82 / 0.39
Consensus (4 LLMs)	RAG agg. · 0.91 / 0.89	RAG agg. · 0.94 / 0.02	GIV-F agg. · 0.85 / 0.40

On YAGO, where 99% of facts are correct, F1 on false facts stays ≤ 0.03 for every strategy: the models call nearly everything true

Shami et al. Benchmarking Large Language Models for Knowledge Graph Validation. EDBT 2026

Before Trusting an LLM Validator

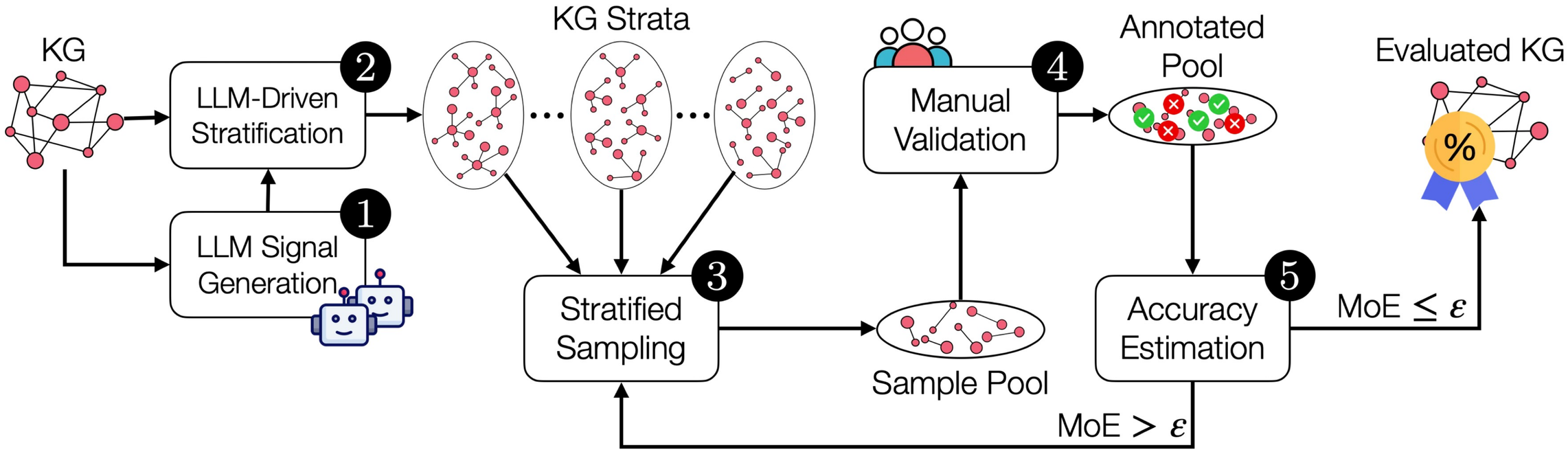
What to test

- Compare with expert labels on your own KG: strong results on semi-synthetic benchmarks do not carry over to real-world facts [1]
- Report class-aware metrics, not plain accuracy: class-wise F1, balanced accuracy, sensitivity and specificity; accuracy hides the minority class (YAGO: F1 on false facts ≈ 0.02) [1–3]
- Estimate KG accuracy from LLM labels on a gold sample and measure the deviation (up to 45%)
- Check compliance, abstentions and how often the model answers “incorrect”: reluctance to do so inflates accuracy on positively skewed datasets [3]
- Test stability across prompts, retrieval and model committees; gains are inconsistent [2]
- Account for cost: RAG is often six or more times slower than direct prompting

[1] Marchesin et al. LLMs as Stratification Signals for KG Accuracy Evaluation. PVLDB 2026 · [2] Shami et al. Benchmarking Large Language Models for Knowledge Graph Validation. EDBT 2026
[3] Marchesin et al. Utility-Oriented KG Accuracy Estimation with Limited Annotations. HCOMP 2024

LLMs as Stratification Signals

LLMs shape the sample, humans label it



The estimator uses human labels only: LLM errors affect efficiency, not unbiasedness

Marchesin et al. LLMs as Stratification Signals for KG Accuracy Evaluation. PVLDB 2026

How to Use LLMs Signals to Build Strata

Soft majority voting over four LLMs

1. **Label:** each LLM judges every triple, correct (1) or incorrect (0) [1]
2. **Triple score:** share of LLMs that judge it correct; values near 0.5 mark disagreement
3. **Cluster score:** mean triple score over the entity cluster
4. **Stratify:** CSRF on cluster scores; zero-variance strata merged with their nearest neighbor [1, 2]
5. **Estimate:** proportional allocation and STWCS; humans label the sample, HPD interval as in 2.2

triple score

$$\mathrm{agg}(t) = \frac{1}{k}\sum_{i=1}^{k} \mathrm{LLM}_i(\mathrm{prompt}(t))$$

cluster score

$$\hat{\mu}_{e\,|\,\mathrm{agg}} = \frac{1}{M_e}\sum_{t \in \mathcal{G}[e]} \mathrm{agg}(t)$$

unbiased estimate

$$\hat{\mu}_{\mathrm{STWCS}} = \sum_{q=1}^{Q} W_q\,\hat{\mu}_q, \qquad W_q = M_q/M$$

k = 4 open LLMs: Cohere Command R, DeepSeek-V3, Llama 3.1 8B, Mistral-Nemo

[1] Marchesin et al. LLMs as Stratification Signals for KG Accuracy Evaluation. PVLDB 2026 · [2] Dalenius and Hodges. Minimum Variance Stratification. J. Amer. Statist. Assoc. 1959

LLM Signals vs Baselines

Annotated triples and cost (hours)

		NELL		DBPEDIA		FactBench	
		μ= 0.91		μ= 0.85		μ= 0.28	
Sampling	Signal	Triples	Cost (hr)	Triples	Cost (hr)	Triples	Cost (hr)
TWCS	none	116±66	1.45±0.83 △	224±85	2.57±0.97 △	252±52	2.89±0.60 △
STWCS	random	109±71	1.36±0.89 △	222±95	2.56±1.10 △	252±64	2.89±0.74 △
STWCS	size	106±75	1.31±0.94 △	218±93	2.52±1.07 △	231±70	2.65±0.81 △
STWCS	llm-agg	**54±40**	**0.67±0.50**	**195±100**	**2.24±1.14**	**200±63**	**2.30±0.72**
STWCS	oracle	42±30	0.51±0.38	46±4	0.52±0.05	102±47	1.18±0.54



LLM-guided stratification cuts annotation cost by **11–54%** over the baselines

llm-agg reduces annotation costs compared to baselines across all datasets, **closing the gap with oracle** stratification

Marchesin et al. LLMs as Stratification Signals for KG Accuracy Evaluation. PVLDB 2026

Scaling Signals by Distillation

Small students trained on LLM labels

- LLMs label 50K triples per KG, 0.25% of a 20M-triple KG; one BERT student per LLM learns the labels, aggregated as distill-llm-agg [1, 2]
- Hard-target distillation uses predicted labels only, unlike soft targets [3], so API-only models suffice [1, 4]
- **Money on YAGO-HQ:** TWCS costs $4,779 in expert labels; with distilled LLM strata, $4,385 of expert labels plus $34 for the LLM labels of the 50K training triples, $4,419 in total. Experts at $5.71 per triple, three per triple [1, 5]

KG (μ)	TWCS	size	distill-llm-agg	oracle
YAGO-HQ (0.75)	3.16	3.20	**2.90**	1.33
YAGO-MQ (0.50)	4.39	4.45	**3.97**	1.99
YAGO-LQ (0.25)	3.31	3.32	**2.83**	1.34

Annotation cost (hours); oracle strata as lower bound [1]

[1] Marchesin et al. LLMs as Stratification Signals for KG Accuracy Evaluation. PVLDB 2026 · [2] Devlin et al. BERT: Pre-training of Deep Bidirectional Transformers. NAACL 2019
[3] Hinton et al. Distilling the Knowledge in a Neural Network. CoRR 2015 · [4] Xu et al. A Survey on Knowledge Distillation of Large Language Models. CoRR 2024 · [5] Paulheim. How much is a Triple? ISWC 2018 (Posters)

Take-Aways: 2.4

Where LLMs stand today

Safe now

- Stratification signals: unbiased estimates, 11–54% lower annotation cost
- Distilled signals for KGs with tens of millions of triples

Premature

- Direct validation: very low F1 on false facts in real-world KGs (YAGO ≈ 0.02, DBpedia ≤ 0.40)
- LLM-derived accuracy: up to 45% off the ground truth

Next: Part 3, putting the estimates to work in entity search

Marchesin et al. LLMs as Stratification Signals for KG Accuracy Evaluation. PVLDB 2026 · Shami et al. Benchmarking Large Language Models for Knowledge Graph Validation. EDBT 2026

Part 3: Downstream Applications and Open Problems

3.1 Veracity-driven entity search

3.2 Open problems and outlook

Collections Used in Part 3

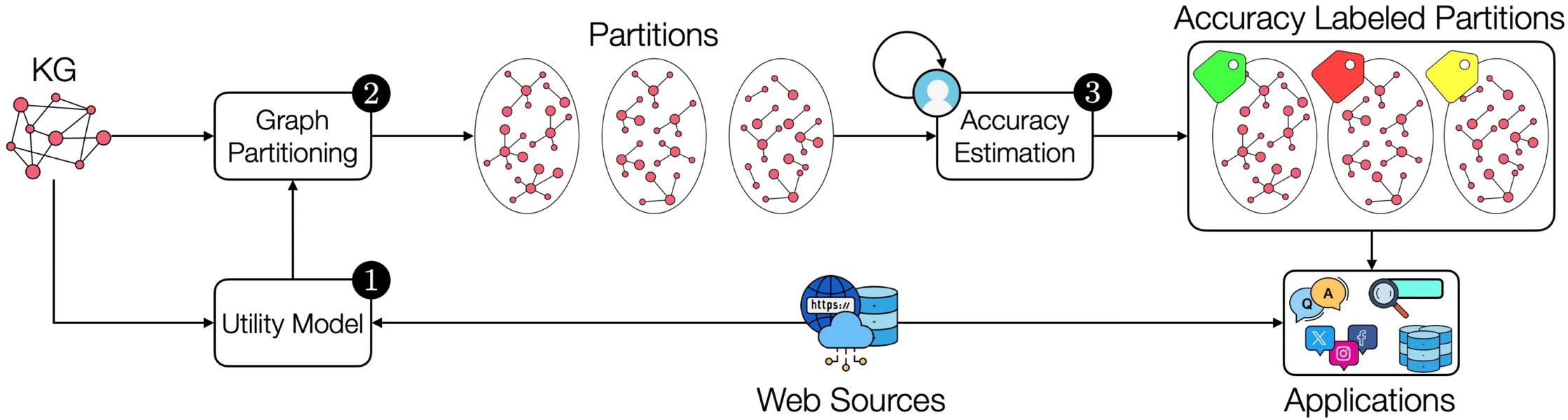
All built on DBpedia 2015-10

Collection	Task in Part 3	Size
DBpedia utility study (2.3)	veracity estimates per partition, consumed by the rankers	7 partitions; 9,930 labeled facts
DBpedia-Entity v2	entity retrieval: rank entities for a query	485 queries; runs of 12 systems
DBpedia-Fact	entity summarization: rank an entity's facts for its card	100 query–entity pairs; 4,069 facts
DBpedia-Defect	defect detection: train and test ALADDIN	1,000 facts held out for test

Marchesin et al. Utility-Oriented KG Accuracy Estimation with Limited Annotations. HCOMP 2024 · Hasibi et al. DBpedia-Entity v2: A Test Collection for Entity Search. SIGIR 2017
Hasibi et al. Dynamic Factual Summaries for Entity Cards. SIGIR 2017 · Irrera et al. Scaling Trust: Veracity-Driven Defect Detection in Entity Search. CIKM 2025

From Estimates to Applications

Veracity estimates as a signal for entity search



Veracity: treated as an alternative term for accuracy in entity search. Partition veracity = fraction of correct facts in a partition, estimated as in Part 2; each fact takes it as its veracity score

Marchesin et al. Veracity Estimation for Entity-Oriented Search with Knowledge Graphs. CIKM 2024

Entity Cards from KG Facts

Summaries pick the KG facts users see



Extracting the most **important facts** about an **entity**

Addressed as a **ranking problem**: rank facts by importance and **pick top-*k*** ones

Balog. Entity-Oriented Search. Springer 2018

Dynamic Entity Summarization

Query-dependent summaries for entity cards



How to generate **query-dependent entity summaries** that can directly address **users' information needs**?



Ranking entity **facts based** on their **importance** for the entity and **relevance** to the query

Einstein family

Born: March 14, 1879, Ulm, Germany
Died: April 18, 1955, Princeton, New Jersey, United States
Siblings: Maja Einstein
Spouse: Elsa Einstein (m. 1919-1936), Mileva Marić (m. 1903-1919)
Education: University of Zurich (1905), more

Einstein awards

Born: March 14, 1879, Ulm, Germany
Died: April 18, 1955, Princeton, New Jersey, United States
Education: University of Zurich (1905), more
Awards: Barnard Medal for Meritorious Service to Science, Academy Award, more
Influenced: Satyendra Nath Bose, Wolfgang Pauli, Tom Cruise, more



Relevance and importance do NOT consider the correctness (veracity) of facts!

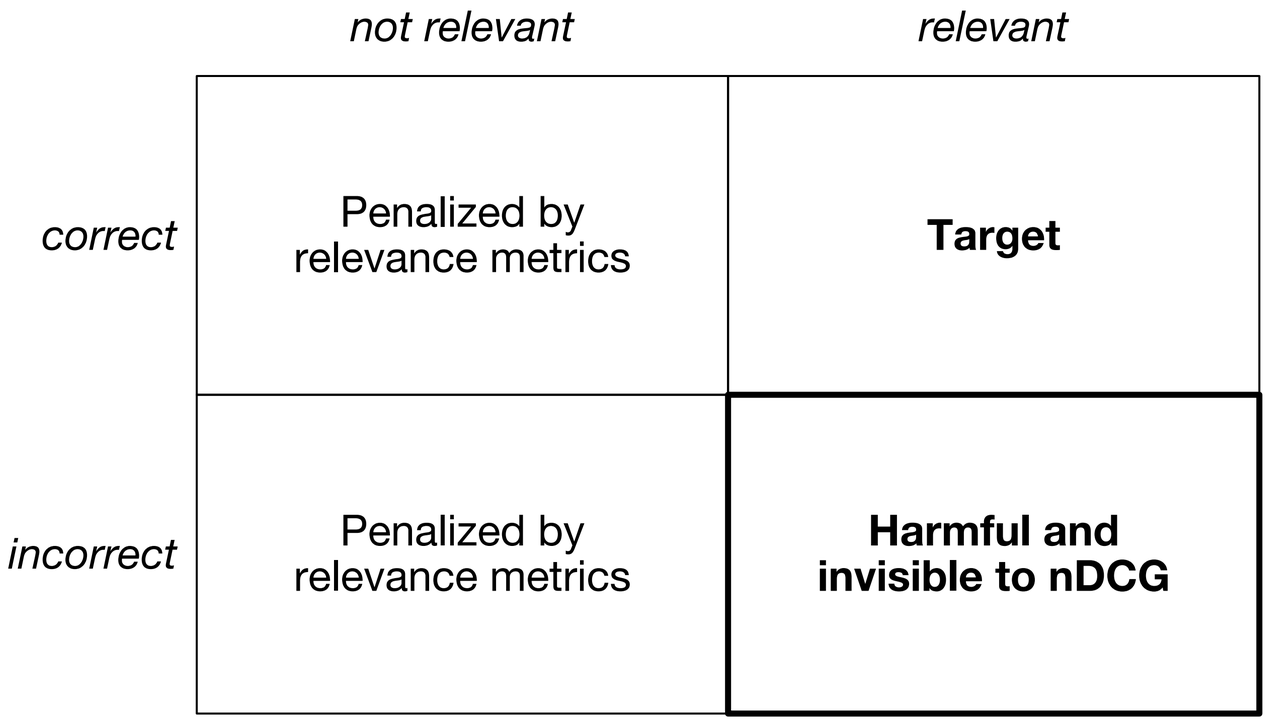


How to increase entity card reliability? **Prioritize** facts with higher (estimated) **quality**

Hasibi et al. Dynamic Factual Summaries for Entity Cards. SIGIR 2017

Relevance and Veracity: Two Axes

What relevance-only evaluation misses



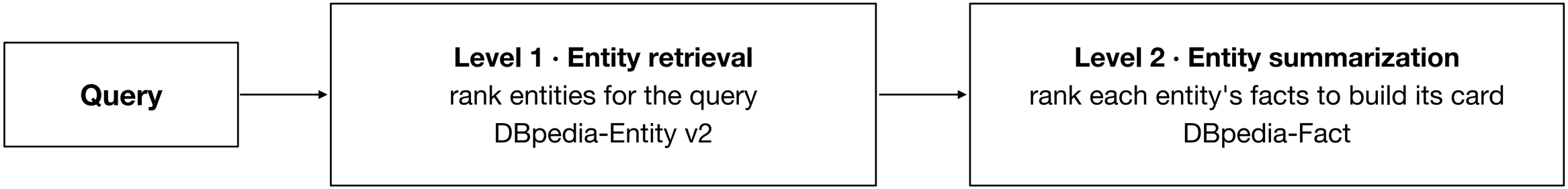
e.g. Einstein, influenced: Tom Cruise

- Entity search benchmarks judge relevance and importance [1, 2] and assume a correct KG [3]
- Misinformation is a cost of search: veracity needs an axis of its own

[1] Hasibi et al. Dynamic Factual Summaries for Entity Cards. SIGIR 2017
[2] Balog. Entity-Oriented Search. Springer 2018
[3] Marchesin et al. Veracity Estimation for Entity-Oriented Search with Knowledge Graphs. CIKM 2024

Two Levels of Entity Search

Rank the entities, then rank their facts



Veracity at level 1: eRank adds an entity's mean fact veracity to its retrieval score

Veracity at level 2: ALADDIN predicts a veracity score for each fact of a card

Both rely on the partition estimates of Part 2 and on human feedback where they disagree

Irrera et al. Scaling Trust: Veracity-Driven Defect Detection in Entity Search. CIKM 2025
Hasibi et al. DBpedia-Entity v2. SIGIR 2017 · Hasibi et al. Dynamic Factual Summaries for Entity Cards. SIGIR 2017

eRank: Entity-Level Veracity

Level 1: veracity in entity ranking

Entity search requires **ranking whole entities**, **not individual facts**, in response to a user query.

$$e\mathrm{Rank}(e) = \mathrm{eScore}(e) + \frac{1}{M_e} \sum_{t \in G[e]} v(t)$$

eRank embeds veracity in the main scoring function: relevance and veracity are combined in a single step.

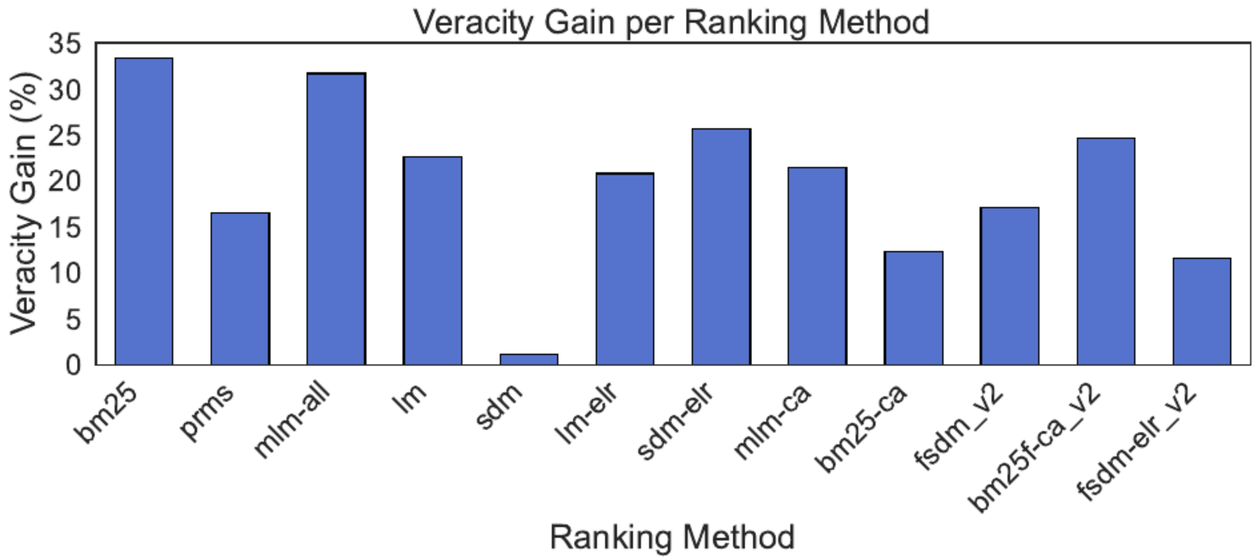
Partitions: the seven DBpedia utility strata of 2.3 (veracity 0.83–0.90)

Irrera et al. Scaling Trust: Veracity-Driven Defect Detection in Entity Search. CIKM 2025

*e*Rank on DBpedia-Entity v2

Official runs of 12 systems [1, 2], before and after *e*Rank

	nDCG@10		nDCG@100	
Model	Orig	*e*Rank	Orig	*e*Rank
BM25	0.26	0.26	0.36	0.37
PRMS	0.39	0.39	0.47	0.47
MLM-All	0.40	0.41	0.49	0.49
LM	0.42	0.42	0.50	0.50
SDM	0.42	0.43	0.51	0.52
LM+ELR	0.42	0.42	0.51	0.51
SDM+ELR	0.44	0.43	0.52	0.52
MLM-CA	0.44	0.44	0.51	0.52
BM25-CA	0.44	0.45	0.53	0.54
FSDM	0.45	0.46	0.53	0.54
BM25F-CA	0.46	0.47	0.55	0.56
FSDM+ELR	0.46	0.46	0.54	0.54



Veracity gain up to 34%; nDCG unchanged (Cohen's d < 0.2)

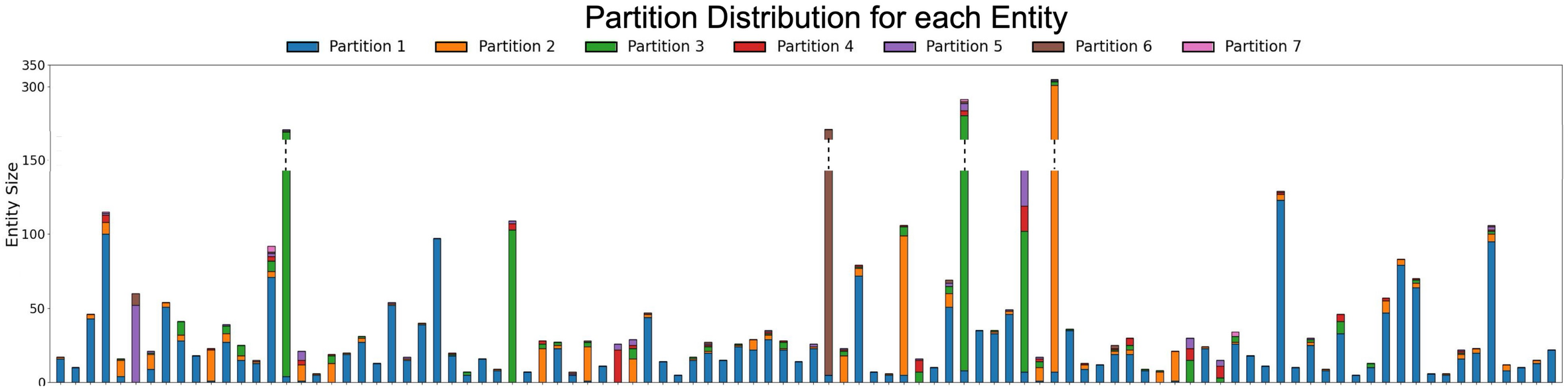
Kendall's $\tau < 0.80$ for all methods at both cutoffs [1]

***e*Rank** promotes entities with higher-veracity facts; relevance is kept

[1] Irrera et al. Scaling Trust: Veracity-Driven Defect Detection in Entity Search. CIKM 2025
[2] Hasibi et al. DBpedia-Entity v2: A Test Collection for Entity Search. SIGIR 2017

Level 2: Fact-Level Veracity

Back to the cards: most facts of an entity share one partition



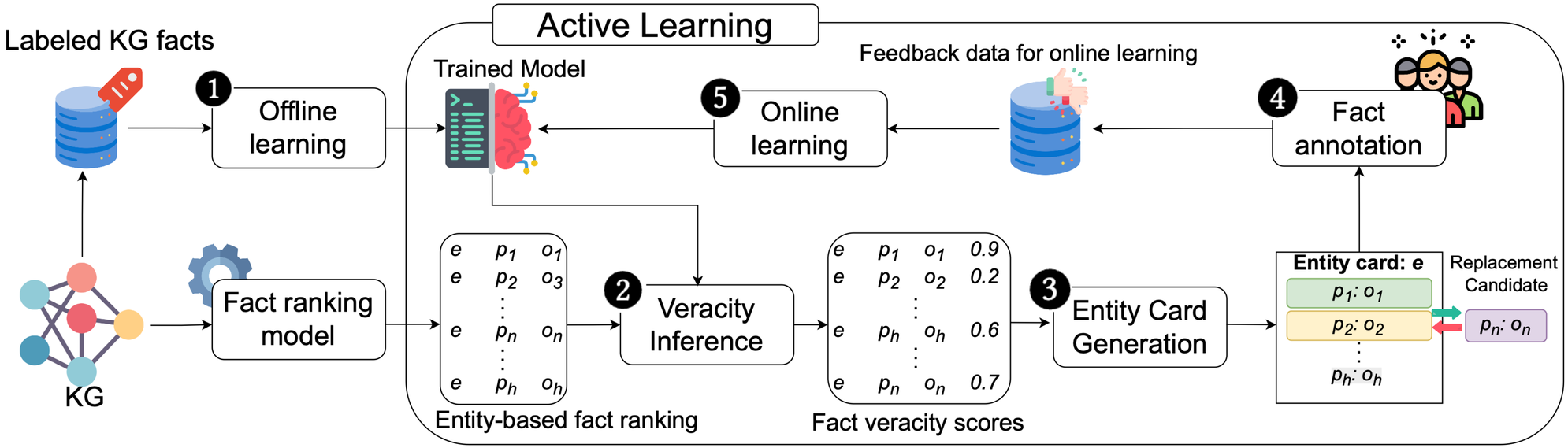
eRank ranks whole entities; a card ranks the facts of one entity. In noisy real-world data, most facts of an entity fall into the same popularity partition, so re-ranking them by partition veracity is ineffective.

Separating true from false facts of the same entity requires finer, fact-level veracity estimation.

Irrera et al. Scaling Trust: Veracity-Driven Defect Detection in Entity Search. CIKM 2025

The ALADDIN System

Two stages, five steps

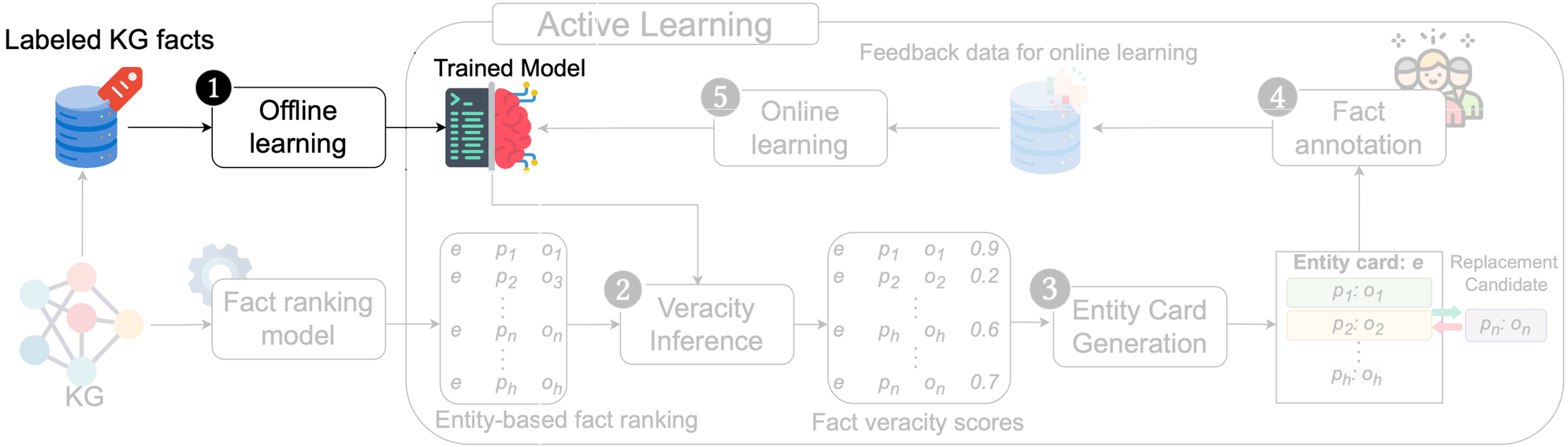


1 offline learning · 2 veracity inference · 3 entity card generation · 4 fact annotation · 5 online learning

Irrera et al. Scaling Trust: Veracity-Driven Defect Detection in Entity Search. CIKM 2025

The ALADDIN System

Offline learning



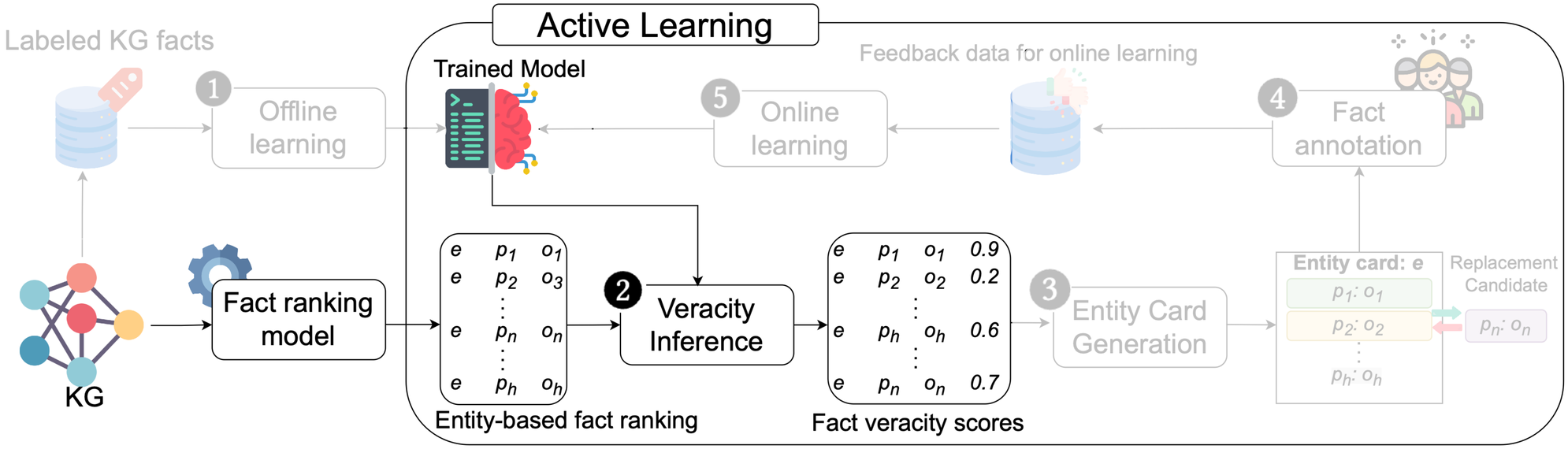
ALADDIN trains a linear regressor on TransE-embedded [2], binary-labeled facts.

It produces a [0,1] veracity score where 0 identifies defects and 1 correct facts. [1]

[1] Irrera et al. Scaling Trust: Veracity-Driven Defect Detection in Entity Search. CIKM 2025
[2] Bordes et al. Translating Embeddings for Modeling Multi-relational Data. NeurIPS 2013

The ALADDIN System

Veracity inference



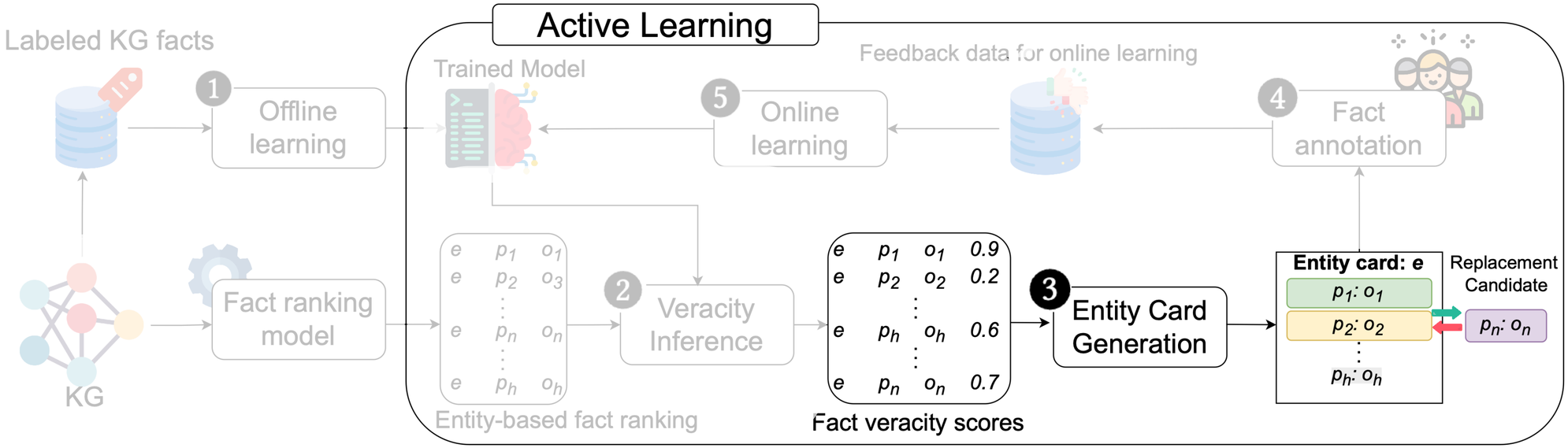
The trained regressor scores every fact in each entity's ranked fact list.

Low-scored yet high-ranked facts are flagged as defects.

Irrera et al. Scaling Trust: Veracity-Driven Defect Detection in Entity Search. CIKM 2025

The ALADDIN System

Entity card generation



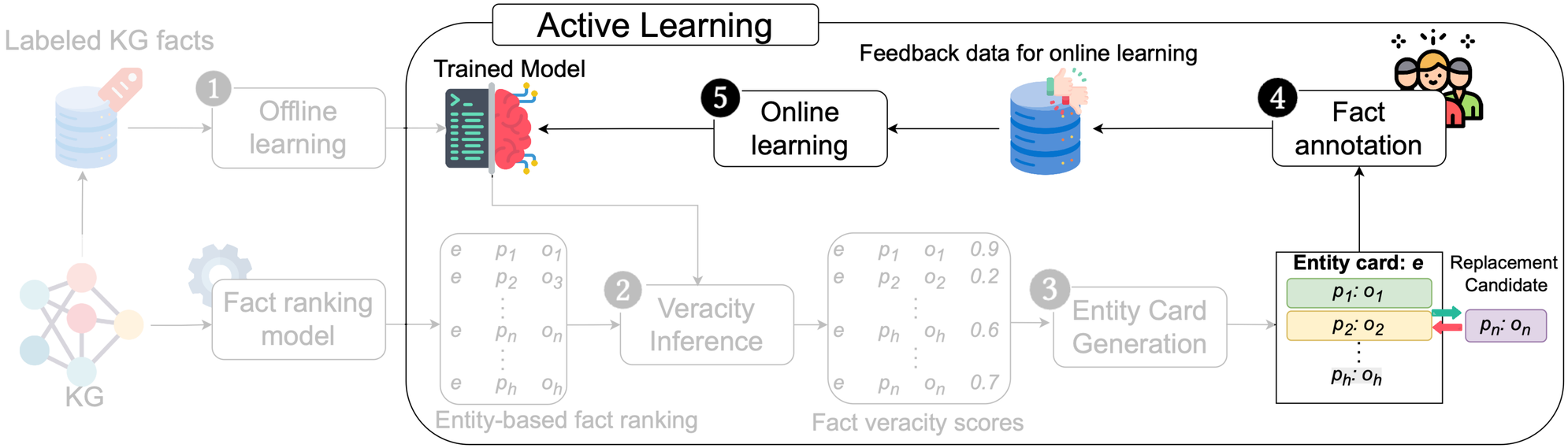
ALADDIN generates an entity card with the top-*h* facts from the fact ranking model.

A threshold θ_v flags low-score facts and proposes higher-veracity replacements from lower ranks, focusing feedback on the most critical, high-visibility facts.

Irrera et al. Scaling Trust: Veracity-Driven Defect Detection in Entity Search. CIKM 2025

The ALADDIN System

Fact annotation and online learning



Annotators review the facts in the entity card. Their feedback powers **incremental (online) updates** of the linear regressor; a residual-based filter keeps only high-disagreement items (Ω_e) to enhance generalization.

$$\Omega_e = \{\langle t_i, \mathbf{1}(t_i)\rangle : |v_{\mathcal{A}}(t_i) - \mathbf{1}(t_i)| > \theta_\delta, i \in [1..h]\}$$

Irrera et al. Scaling Trust: Veracity-Driven Defect Detection in Entity Search. CIKM 2025

ALADDIN: Defect Detection

Online vs offline ALADDIN on DBpedia-Defect

	binary F1	weighted F1	balanced accuracy
$\mathcal{A}_{\text{off}}$	0.79	0.70	0.63
$\mathcal{A}_{\text{on}}$	0.83	0.72	0.64



Online ALADDIN ($\mathcal{A}_{on}$) vs its offline version without active learning ($\mathcal{A}_{off}$), on 1,000 held-out DBpedia-Defect facts



binary F1 rises **0.79→0.83**, with additional gains in **weighted F1** and **balanced accuracy**, confirming the benefit of active online learning in defect detection.

Irrera et al. Scaling Trust: Veracity-Driven Defect Detection in Entity Search. CIKM 2025

ALADDIN: Defect Recommendation

Precision and recall when inspecting 10% to 50% of the facts

	Precision			Recall		
Budget	$\mathcal{A}_{\text{off}}$	$\mathcal{A}_{\text{on}}$	% gain	$\mathcal{A}_{\text{off}}$	$\mathcal{A}_{\text{on}}$	% gain
10%	0.42	0.42	–	0.65	0.65	–
25%	0.29	0.66	+127.58	0.54	0.63	+16.66
50%	0.61	0.66	+8.19	0.58	0.66	+13.80



Online ALADDIN vs its offline version: similar at a 10% budget; at 25%, precision rises from 0.29 to 0.66 (+127.58%)

Irrera et al. Scaling Trust: Veracity-Driven Defect Detection in Entity Search. CIKM 2025

ALADDIN: Summaries and Cards

$\mathcal{A}$Rank = facts re-ranked with ALADDIN veracity scores

Model	nDCG@5			nDCG@10		
	Orig	vRank	$\mathcal{A}$Rank	Orig	vRank	$\mathcal{A}$Rank
RELIN	0.47	0.50	0.46	0.54	0.56	0.53
DynES/i	0.72	0.73	0.70	0.76	0.76	0.75
DyneES/r	0.62	0.62	0.60	0.66	0.66	0.64
DynES/u	0.76	0.76	0.74	0.79	0.79	0.77



$\tau \geq$ **0.79** and $\tau_{AP} \geq$ **0.77** with vRank (re-ranking by stratum veracity)



$\tau \leq$ **0.28** and $\tau_{AP} \leq$ **0.26** with $\mathcal{A}$Rank



vRank mostly preserves the original order. ALADDIN reshuffles the ranking order to prioritize higher-veracity facts without affecting effectiveness.



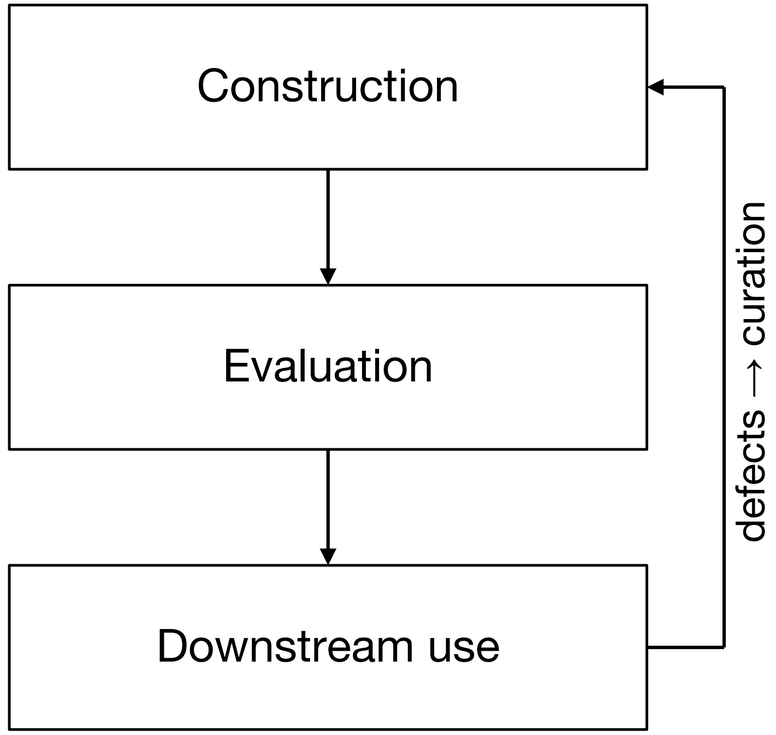
ALADDIN's reranking improves perceived entity card quality.
ALADDIN cards were preferred **40%** vs **26%** originals (ties **34%**), i.e., **better or equal in 74%** of cases.

Irrera et al. Scaling Trust: Veracity-Driven Defect Detection in Entity Search. CIKM 2025

Closing the Loop

From detected defects back to construction

- Stratum estimates from Part 2 are the veracity that eRank consumes [1]
- ALADDIN learns only from annotations where model and human disagree (residual > 0.5)
- ALADDIN asks for labels where the model is unsure, not at random: good for finding defects, not for estimating accuracy, which still needs the random samples of Part 2
- Flagged defects can feed curation and retraining, as in CORE's active learning workflow (1.1); this integration is left to future work



[1] Irrera et al. Scaling Trust: Veracity-Driven Defect Detection in Entity Search. CIKM 2025
[2] Marchesin et al. Building a Large Gene Expression-Cancer Knowledge Base with Limited Human Annotations. Database 2023

Take-Aways: 3.1

Veracity-driven entity search



Entity rankings can absorb veracity without losing relevance



Partition veracity is too coarse within an entity: score each fact



A linear model w/ targeted human feedback finds defects under small budgets



Defect detection and accuracy estimation answer different questions: use both

Next: what remains open (3.2)

Irrera et al. Scaling Trust: Veracity-Driven Defect Detection in Entity Search. CIKM 2025

Part 3: Downstream Applications and Open Problems

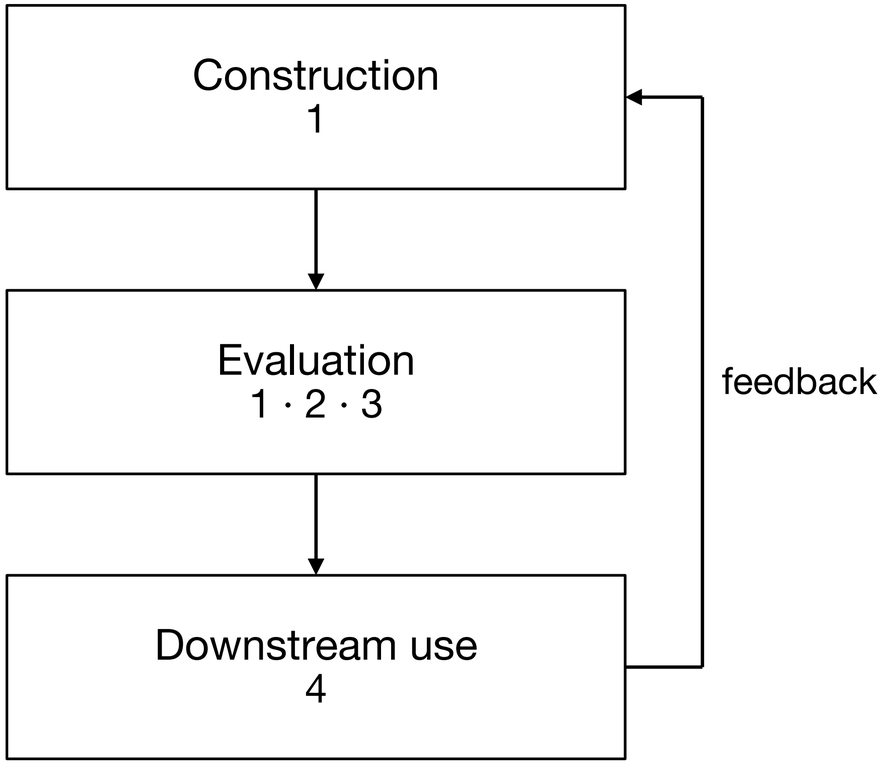
3.1 Veracity-driven entity search

3.2 Open problems and outlook

Open Problems

Four directions along the lifecycle

1. Evolving KGs: requirements, challenges, temporal dynamics
2. Transfer of signals and models across KGs
3. Soft-label annotation
4. KG accuracy for LLM-based applications



The Source-Centric View

How KG quality is usually managed

- Select reliable sources, estimate their trustworthiness, resolve conflicts: data fusion, truth discovery [1–3]
- Document and govern ingestion pipelines, record provenance [4, 5]
- Quality evaluation: evidence-bearing measurement of fitness-for-use properties [6]
- Instruments: constraint validation (SHACL), sampling-based manual verification, provenance audits [7, 8]
- Recent work questions this view for KGs in dynamic, evolving domains [9, 10]

[1] Dong et al. Integrating Conflicting Data: The Role of Source Dependence. PVLDB 2009 · [2] Dong et al. Knowledge-Based Trust. PVLDB 2015
[3] Li et al. A Survey on Truth Discovery. SIGKDD Explor. 2015 · [4] Paulheim. Knowledge Graph Refinement: A Survey. Semantic Web 2017
[5] Xue and Zou. Knowledge Graph Quality Management: A Comprehensive Survey. IEEE TKDE 2023 · [6] Zaveri et al. Quality Assessment for Linked Data: A Survey. Semantic Web 2016
[7] Gao et al. Efficient Knowledge Graph Accuracy Evaluation. PVLDB 2019
[8] Marchesin et al. Cost-Effective Quality Evaluation for Evolving Knowledge Graphs. ACM J. Data Inf. Qual. 2026
[9] Mohammed et al. The Five Facets of Data Quality Assessment. SIGMOD Rec. 2025 · [10] Polleres et al. How Does Knowledge Evolve in Open Knowledge Graphs? TGDK 2023

Three Requirements

What evolving KGs must support

Requirement	Question	Example
R1 Temporal provenance and history	What does the KG record?	Validity interval, evidence, and prior states of 'gene X is a tumor suppressor'
R2 Context-dependent correctness	How is correctness assessed?	plays-for(A, Team1) judged true for 2022 and false for 2024
R3 Cost-aware quality management	How often, how thoroughly?	Which acquisition updates are re-verified now and which later

R1 is about data, R2 about judgment, R3 about budget; R2 presupposes R1

Marchesin et al. Cost-Effective Quality Evaluation for Evolving Knowledge Graphs. ACM J. Data Inf. Qual. 2026

Four Facets of Evolving-KG Quality

Why source quality does not guarantee KG quality

Origin of issues

Source: sources change, disagree, drift; a paper retracted in 2024 was the evidence for a 2019 assertion (E1)

Reliability: accepted facts decay after updates; an ingestion-time check misses the 2024 transfer (E2) [2]

Temporal: assertions hold within validity intervals (E1, E2, E3)

Constraint on the response

Utility: not every assertion can be re-verified; thousands of dependent updates share one budget (E3)

R1 records along source, temporal, reliability; R2 reads the record to judge; R3 imposes utility [1]

[1] Marchesin et al. Cost-Effective Quality Evaluation for Evolving Knowledge Graphs. ACM J. Data Inf. Qual. 2026
[2] Polleres et al. How Does Knowledge Evolve in Open Knowledge Graphs? TGDK 2023

Open Challenges for Evolving KGs

Five challenges from the requirements

Challenge [1]	Requirements
C1 Data and metadata management at scale: provenance, versions, snapshot-delta storage	R1, R3
C2 Heterogeneous temporal semantics and update regimes	R1, R2
C3 Representation and querying of time-stamped, evidence-carrying statements	R1, R2
C4 Limits of automated reliability testing: SHACL plus temporal coherence [2]	R2
C5 Change-rate classification: what to re-verify, and when	R3

[1] Marchesin et al. Cost-Effective Quality Evaluation for Evolving Knowledge Graphs. ACM J. Data Inf. Qual. 2026
[2] Polleres et al. How Does Knowledge Evolve in Open Knowledge Graphs? TGDK 2023

Temporal Dynamics

Keeping estimates valid as the KG changes

- Open KGs change continuously, and their edit histories record how [1]
- Incremental designs handle append-only updates and treat past labels as still valid [2]
- **Open:** detect when a stored label expires (R2: correctness depends on time) and decide what to re-verify (C5: change-rate classification) under budget [3]
- **Open:** carry posteriors across releases as priors without inheriting stale evidence [4]

[1] Polleres et al. How Does Knowledge Evolve in Open Knowledge Graphs? TGDK 2023
[2] Gao et al. Efficient Knowledge Graph Accuracy Evaluation. PVLDB 2019
[3] Marchesin et al. Cost-Effective Quality Evaluation for Evolving Knowledge Graphs. ACM J. Data Inf. Qual. 2026
[4] Marchesin and Silvello. Credible Intervals for Knowledge Graph Accuracy Estimation. Proc. ACM Manag. Data (SIGMOD) 2025

Transfer Across KGs

Do signals and students generalize?

- Students are distilled per KG, from LLM labels on ≤ 0.25% of its triples [1]
- LLM validators: strong on semi-synthetic benchmarks, weak on real-world facts
- **Open:** train once, stratify many KGs; detect when a transferred signal degrades
- Low risk for validity: a poor signal costs efficiency, not unbiasedness
- *Related:* knowledge distillation of LLMs [2]

[1] Marchesin et al. LLMs as Stratification Signals for KG Accuracy Evaluation. PVLDB 2026
[2] Xu et al. A Survey on Knowledge Distillation of Large Language Models. CoRR 2024

Soft Labels

Cheaper judgments, valid estimates?

- Binary labels plus IDK are costly and still disputed: experts agreed on 77% of DBpedia facts, $\kappa = 0.51$ [1]
- Aggregation: majority voting, reliability-weighted voting, learned aggregators [1–3]
- **Open:** graded or probabilistic labels whose uncertainty propagates into the interval
- **Open:** guarantees when labels mix experts, crowd workers and models
- **Open:** cheap probabilistic judges, e.g. non-generative “System One” models such as Jev: soft labels or strata at a fraction of LLM cost? Untested on KG facts [4, 5]

[1] Marchesin et al. Utility-Oriented KG Accuracy Estimation with Limited Annotations: A Case Study on DBpedia. HCOMP 2024
[2] Hung et al. An Evaluation of Aggregation Techniques in Crowdsourcing. WISE 2013
[3] Zhang et al. Learning from Crowdsourced Labeled Data: A Survey. Artif. Intell. Rev. 2016
[4] Vals AI. Independent Evaluation of Jev. 2026 (blog) · [5] Denney and DiGiuseppe. Jev versus LLMs: Accuracy, Cost and Calibration. arXiv:2610.06625

KG Accuracy for LLM Applications

Grounding needs evidence of its own quality

- LLM answers grounded in KG facts inherit the KG's errors [1]
- Long-form factuality splits outputs into atomic facts and verifies each: the same estimation problem at answer level [2]
- **Open:** propagate KG uncertainty into answer-level confidence; measure hallucination rates with intervals

[1] Huang et al. A Survey on Hallucination in Large Language Models. ACM Trans. Inf. Syst. 2025
[2] Wei et al. Long-form Factuality in Large Language Models. NeurIPS 2024

Take-Home Messages

One message per lifecycle stage, plus LLMs



Construction choices decide where errors enter; facts keep changing after deployment

Marchesin et al., Database 2023; Marchesin et al. Cost-Effective Quality Evaluation for Evolving Knowledge Graphs. ACM J. Data Inf. Qual. 2026



Estimate accuracy with sampling designs that match the annotation cost

Gao et al., PVLDB 2019



Report intervals that do not collapse: Wilson confidence intervals or HPD credible intervals

Marchesin and Silvello, PVLDB 2024; Proc. ACM Manag. Data (SIGMOD) 2025



Prioritize by utility as a design choice

Marchesin et al., HCOMP 2024



Use LLMs as stratification signals, not as direct validators (yet)

Shami et al., EDBT 2026; Marchesin et al., PVLDB 2026



Downstream systems need veracity next to relevance, at entity and fact level

Marchesin et al., CIKM 2024; Irrera et al., CIKM 2025

Datasets at a Glance (1/2)

Benchmarks for KG accuracy evaluation

Dataset	Facts	Clusters	Accuracy μ	Labels	Papers	Tutorial
YAGO	1,386	822	0.99	crowd	PVLDB 2019, 2024 · SIGMOD 2025 · EDBT 2026	2.1, 2.2, 2.4
NELL	1,860	817	0.91	crowd	PVLDB 2019, 2024, 2026 · SIGMOD 2025	2.1, 2.2, 2.4
DBpedia 2015-10	9,344	2,936	0.85	crowd + expert QC	HCOMP 2024 · SIGMOD 2025 · EDBT 2026 · PVLDB 2026 · CIKM 2025	1.2, 2.2–2.4, 3.1
FactBench	2,800 / 4,219	1,157	0.54 / 0.28	synthetic errors	SIGMOD 2025, EDBT 2026 / PVLDB 2026	2.2, 2.4
DisGeNET	2,999,087	21,243	0.76 (CEM) 0.9 / 0.5 / 0.1 (TEM)	synthetic	PVLDB 2024	2.2
SYN 100M	101,415,011	5,000,000	0.9 / 0.5 / 0.1	synthetic	PVLDB 2024 · SIGMOD 2025	not shown
YAGO-HQ / MQ / LQ	20,433,311	4.2–4.4M	0.75 / 0.50 / 0.25	synthetic errors	PVLDB 2026	2.4

Clusters = entity clusters. PVLDB 2019 = Gao et al., who also use MOVIE, a KG built at Amazon that is not public. DisGeNET and SYN 100M carry synthetic labels whose accuracy is set by the label model (CEM: cluster error model; TEM: triple error model with error rates 0.1, 0.5, 0.9).

Ojha and Talukdar. KGEval. EMNLP 2017 (YAGO, NELL) · Marchesin et al. Utility-Oriented KG Accuracy Estimation with Limited Annotations. HCOMP 2024 (DBpedia)
Gerber et al. DeFacto. J. Web Semant. 2015 (FactBench) · Piñero et al. The DisGeNET Knowledge Platform for Disease Genomics. Nucleic Acids Res. 2020
Suchanek et al. YAGO 4.5. SIGIR 2024

Datasets at a Glance (2/2)

Construction and downstream collections

Collection	Task	Size	Papers	Tutorial
CORE knowledge base	KG construction from the literature (gene expression and cancer)	153,016 facts (KG3)	Database 2023	1.1
DBpedia utility study	utility-oriented accuracy estimation; utility from LSQ 2.0 SPARQL logs	37,546 labels on 11,296 facts	HCOMP 2024	1.2, 2.3
FactCheck RAG corpus	LLM fact validation with retrieved web evidence	2,090,305 documents for 13,530 triples	EDBT 2026	2.4
DBpedia-Entity v2	entity retrieval: ranking entities for a query (eRank)	485 queries, 4.6M entities, runs of 12 systems	CIKM 2025	3.1
DBpedia-Fact	entity summarization: ranking an entity's facts for its card	100 query–entity pairs, 4,069 facts	CIKM 2024, 2025	3.1
DBpedia-Defect	fact-level defect detection: training and testing ALADDIN	utility-study facts; 1,000 held out for test	CIKM 2025	3.1

Hasibi et al. DBpedia-Entity v2: A Test Collection for Entity Search. SIGIR 2017 · Hasibi et al. Dynamic Factual Summaries for Entity Cards. SIGIR 2017 (DBpedia-Fact)
Stadler et al. LSQ 2.0. Semantic Web 2024

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Reading list by part; alphabetical within each part

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D. Esteves et al. Toward Veracity Assessment in RDF Knowledge Bases: An Exploratory Analysis. ACM J. Data Inf. Qual. 2018.

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Questions?